Harmonic-Tail Reconstruction, Quartic Trigamma Moments, and Spectral Stieltjes Calculus

Exact continuations of the ProveIt polylogarithm programme

Research continuation prepared for Vladimir Reshetnikov

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Abstract

We continue several specific problems in the ProveIt manuscript and its incoming reports. First, we complete optimal-prefix reconstruction for every unequal even pair with smaller tail order eight. A structural factorization identifies the highest Bernoulli-degree determinant with a strictly positive zeta moment determinant; an exact finite certificate closes the residue class left open in the preceding report. The same factorization proves effective eventual reconstruction for every fixed smaller even order, reducing each fixed-order conjecture to finitely many explicit rational matrices.

Second, we evaluate every regular centered quartic trigamma moment after prescribed polynomial subtraction. The values lie in six specified formal coordinates, and a finite Bernoulli transform, an anchored analytic generator, and a finite polylogarithm construction of depth at most three are given. Hoffman's classical unweighted quartic evaluation supplies one starting constant and is explicitly credited.

Third, we evaluate the second spectral factorization defect for any number of quadratic masses by ordinary logarithmic sums and an elementary one-dimensional resolvent integral, including every confluent coefficient. Finite formulas at all higher spectral derivatives retain the otherwise lost Hurwitz jets. A one-sided counterpart identifies all generalized Stieltjes corrections, treats the source's conjugate-linear factorization problem, and gives an anchored dilogarithm–Gamma primitive. We also correct a research-status overlap concerning the already established entire multiple-harmonic interpolant. Proofs, exact finite certificates, independent numerical diagnostics, provenance, and a concrete further-research agenda accompany the article.

Source snapshot: `73f18ef013d5b74eafc883efb2c15210642df12a`.
All claims of continuation are relative to this inspected snapshot; the article is a research draft with explicit proofs and replay artifacts.

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1 Scope, exact targets, and conventions

The useful unit of progress in this programme is a proved identity or a precisely delimited structural theorem. Several nearby special-value problems already have substantial proofs, and several plausible-looking new directions are already present in earlier reports. Accordingly, the results below are attached to specific source questions before their proofs are developed. Bounds appear only inside the reconstruction and convergence proofs; they are not independent objectives of the article.

The principal new conclusions are the complete order-eight theorem, the effective fixed-order theorem, the all-moment quartic evaluation, and the higher spectral factorization calculus. These conclusions have different logical content. A nonzero rational determinant proves a coefficient-rank statement; it does not prove independence of the real constants later substituted into the coordinates. A normally convergent subtracted sum is an ordinary identity; the divergent raw sum preceding that subtraction is not assigned an unspecified value.

1.1 The concrete questions being continued

Source target	Result here	Boundary retained
Reconstruction report, Q1: the omitted class $b \equiv 1 \pmod{16}$ for $(8, 2b)$ [2]	Theorem 2.1 covers every $b \geq 5$.	The unrelated Gaussian period conjectures are unaffected.
Reconstruction report, Q2: general unequal even orders	Theorem 3.1 gives a terminating threshold algorithm for each fixed smaller order.	The remaining finite set is not checked for every smaller order.
Reconstruction report, Q5: a quartic formula, fixed space, and generator	Theorems 4.2 and 4.3 give the complete regular parity family and six formal coordinates.	Fifth and higher powers require additional analysis.
Quadratic-spectrum report, Q7: several factors beyond the first derivative [5]	Theorems 5.1, 5.2, and 5.3 give the cross term and its full confluent series.	A finite classical-polylogarithm reduction of the integral at arbitrary parameters is not asserted.
Quadratic-spectrum report, Q2: higher conjugate-linear defects	Corollary 6.4 gives every spectral order, with the correct scale and all Stieltjes terms.	The first-derivative parity criterion does not eliminate higher nonlocal terms.
Same-direction report, Q2: entire higher-depth connected interpolation [6]	Section 7 identifies the existing all-depth interpolant and geometric kernel exactly.	This is an attribution and research-status correction, not a new entire-family theorem.

Two additional constructions go beyond those literal questions. Theorem 6.1 isolates the Hurwitz-pole contribution at every derivative order for arbitrary polynomial weights and several quadratic factors. Theorem 6.5 integrates a nonlinear second-jet identity explicitly in the shift, using dilogarithms and a Gamma ratio. Both have their normalization fixed in the statement.

1.2 Source pin and attribution

The repository was inspected at commit `73f18ef013d5b74eafc883efb2c15210642df12a`, including the canonical manuscript status files and the relevant harmonic, Gaussian, Hurwitz, and spectral branches. All six archives currently in `docs/incoming` at that pin were extracted and triaged. The detailed proof development uses the reconstruction, quadratic-spectrum, and

same-direction reports; the Beta-crossing, Gamma-continuation, and shift-and-twist reports were also considered when selecting targets. The exact source paths and SHA-256 digests are recorded in `provenance.json`. This is not an assertion that every theorem in the four-volume manuscript has been audited.

Classical ingredients include Bernoulli–Hurwitz asymptotics, Euler's Bernoulli formula, polygamma partial fractions, Gamma reflection, and normalized Mellin continuation [13]. Hoffman's quartic tail sum [9], the Ihara–Nakamura–Yamamoto interpolant [10], and the first-derivative multiplicative-anomaly framework [11, 12] are attributed where used. Inherited determinant reductions and cubic formulas are identified explicitly. No claim of exhaustive historical priority is inferred from absence in the inspected repository.

1.3 Common conventions

We use

$$H_n^{(s)} = \sum_{k=1}^{n} k^{-s}, \qquad H_0^{(s)} = 0, \qquad A_s(n) = \zeta(s, n+1), \qquad x_n = n + \tfrac{1}{2}$$

in the centered harmonic sections. For an integer $s \geq 2$, $A_s(n) = \zeta(s) - H_n^{(s)}$. The polygamma convention is

$$\psi^{(m)}(z) = (-1)^{m+1} m!\, \zeta(m+1, z), \qquad m \geq 1.$$

In particular $A_2(n) = \psi'(n+1)$ and $A_{2k}(n) = \psi^{(2k-1)}(n+1)/(2k-1)!$.

Bernoulli polynomials are normalized by

$$\frac{te^{xt}}{e^t - 1} = \sum_{m \geq 0} B_m(x) \frac{t^m}{m!}, \qquad B_m = B_m(0), \qquad B_1 = -\tfrac{1}{2}.$$

The crucial centered zero is

$$\zeta(-2j, 1/2) = 0 \qquad (j \geq 0).$$

It makes the specified polynomial subtractions have zero continued sum. A shifted analogue must restore its nonzero Hurwitz values; this midpoint normalization is not silently transported to other shifts.

Superscripts on ζ and on spectral families denote derivatives in their first, spectral variable unless stated otherwise. We use

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(a) u^m, \qquad \gamma_0(a) = -\psi(a).$$

Thus the order derivative of ζ and the parameter derivative of a Stieltjes constant are different operations. All exchanges of spectral differentiation and infinite sums below follow a locally uniform subtraction argument, rather than formal differentiation at a pole.

The one-variable multiple polylogarithm uses the outer-index argument:

$$\mathrm{Li}_{s_1,\dots,s_d}(z) = \sum_{n_1 > \cdots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \cdots n_d^{s_d}}.$$

At $z = 1$ with admissible positive integer indices this is the descending-index multiple zeta value. In the finite generator formulas we first take $z = e^{-t}$ with real $t > 0$; the anchored integral formulas specify the holomorphic continuation through the cancellations at $t = 0$.

1.4 Why the proofs use subtractions and leading determinants

The harmonic and spectral problems share a finite algebraic step but have different analytic mechanisms. For reconstruction, the source matrix is indefinite. Its highest degree in a fixed number of Bernoulli variables nevertheless factors into a positive discrete moment determinant. The factorial size of the Bernoulli variables makes that highest-degree part dominate eventually. The exact exceptional arithmetic remains visible and is handled separately for order eight.

For quartic moments, summation by parts reduces the problem from the fourth tail power to the inherited cubic coordinates. A specific boundary cancellation removes the remaining cubic Bernoulli convolution. The surviving lower-power terms explain all exceptional low indices; they cannot be discarded by parity alone.

For spectral derivatives, the common-scale difference vanishes to order two in the spectral parameter before summation. A resonant Hurwitz pole can lower that order by one. Its residue, regular Stieltjes coefficients, and higher coefficient of the vanishing factor all matter. The finite subtraction formulas retain those contributions while leaving an ordinary summable remainder. This is the source of the extra local term in (6.22) and of the higher bilateral correction in (5.34).

2 The complete order-eight reconstruction theorem

The incoming reconstruction report [2, Section 5] proves optimal-prefix reconstruction for $(p,q)=(8,2b)$ when $b\not\equiv 1 \pmod{16}$, while explicitly leaving the remaining residue class open. We prove nonvanishing for that class as well. The proof identifies the highest Bernoulli-degree part of the exact four-dimensional determinant as a positive zeta moment determinant. A uniform comparison handles all $b\ge 512$; exact finite arithmetic then closes the 31 outstanding smaller cases. The conclusion is an exact reconstruction theorem and yields ordinary convergent identities for products of higher polygammas. No arithmetic independence of special values is used or claimed.

2.1 Source coordinates and the matrix whose rank is at issue

Put

$$A_s(n)=\zeta(s,n+1)=\zeta(s)-H_n^{(s)},\qquad x_n=n+\tfrac12,\qquad D_{b,r}=D_{8,2b}(-2r),$$

where

$$D_{8,2b}(s)=\sum_{n\ge 0}A_8(n)A_{2b}(n)x_n^{-s}$$

is initially defined for $\operatorname{Re} s>-2b-5$ and is continued by its centered asymptotic expansion. Its nonpositive even arguments are regular. The inherited inverse Bernoulli coordinates are

$$C_j=\sum_{r=0}^{j}\binom{2j+1}{2r}2^{-2(j-r)}D_{b,r}, \tag{2.1}$$

$$D_{b,r}=\sum_{j=0}^{r}\frac{\binom{2r}{2j}}{2j+1}B_{2r-2j}(1/2)C_j. \tag{2.2}$$

For $j\ge b$ these reduce to a rational linear combination of $Z_k=\zeta(2k)$, $0\le k<b$, where $Z_0=-1/2$. The exact late block $M_b=(M_{ik})_{0\le i,k<b}$ is

$$M_{ik}=B_{2i+2k-6}\left\{\frac{\binom{2i+2}{8-2k}}{2i+2}+\frac{\binom{2b+2i-6}{2b-2k}}{2b+2i-6}\right\}. \tag{2.3}$$

An entry with negative Bernoulli index is zero. A binomial summand is omitted unless its lower index is at least 2 and at most its upper index. This exclusion includes lower index zero. Thus

$$(C_b, \dots, C_{2b-1})^{\mathsf{T}} = M_b (Z_0, \dots, Z_{b-1})^{\mathsf{T}}.$$

More generally, use the same entry formula for every $i \geq 0$ and $0 \leq k < b$. The finite Bernoulli-binomial moment formula from [3] is

$$C_j = \sum_{k=0}^{b-1} M_{j-b,k} Z_k \qquad (j \geq b).$$

Thus these coefficients also specify every later coordinate, not only the square block used for the rank proof. The determinant of the first $2b$ moments in the formal coordinate list $C_0, \dots, C_{b-1}, Z_0, \dots, Z_{b-1}$ is $\det M_b/(4b-1)!!$. This is a statement about rational coefficient rank, before evaluating the coordinates as complex numbers.

Theorem 2.1 (All unequal pairs with smaller order eight)**.** *For every integer $b \geq 5$, $\det M_b \neq 0$. Consequently, for every even $q > 8$, the moments*

$$D_{8,q}(0), D_{8,q}(-2), \dots, D_{8,q}(-2(q-1))$$

have full formal rank q, and every later centered even moment is an explicit rational linear combination of this prefix.

2.2 An explicitly defined four-dimensional determinant

We retain the source normalization, to make the comparison of claims and certificates unambiguous. For $0 \leq i, \ell \leq 3$ set

$$n_i = b - 4 + i, \quad N_{i\ell} = 2(n_i - \ell) + 1, \quad R_\ell = 2b - 6 + 2\ell,$$

$$F_{i\ell} = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n_i - r)(2n_i + 3 + r).$$

An empty product is one. For

$$W_{i\ell}(z) = (z + 2\ell + 1)(2n_i + 2\ell + z + 1)^{\underline{2i}},$$

define the Newton coefficients

$$d_{i\ell h} = \frac{1}{h!} \sum_{j=0}^{h} (-1)^{h-j} \binom{h}{j} W_{i\ell}(N_{i\ell} - j). \tag{2.4}$$

The underline means a falling factorial. With independent variables $U_1, \dots, U_7$, put

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)! N_{i\ell}!} \sum_{\substack{1 \leq h \leq 2i+1 \\ h \text{ odd}}} d_{i\ell h} N_{i\ell}^{\underline{h}} U_{4-i+\ell+(h-1)/2}. \tag{2.5}$$

Ratios of factorials with fixed index difference mean their finite rational products in b.

Set

$$\begin{aligned} D_8(b) = {} & 256b^8 - 3584b^7 + 18496b^6 - 40544b^5 + 25984b^4 \\ & + 18424b^3 - 5856b^2 - 8316b - 2025, \end{aligned} \tag{2.6}$$

$$P(b;U) = 576(2b-5) D_8(b) \frac{\det \mathcal{A}(b;U)}{\det F(b)}. \tag{2.7}$$

The source's finite polynomial certificate proves $P \in \mathbb{Z}[b, U_1, \dots, U_7]$, of degree at most four in U, and gives

$$\begin{aligned}\det F = \frac{256}{4725}&(b-2)(b-1)(2b-5)(2b-3)(4b-11)(4b-9)\\ &\cdot (4b-7)^2(4b-5)(4b-3)D_8(b).\end{aligned} \tag{2.8}$$

The integer $D_8(b)$ is odd at every integer b, so these denominators are nonzero in the stated range.

Here is its exact connection to the source matrix. Let

$$\mu_r = \begin{cases} B_{2r}/(2r)!, & r \geq 0,\\ 0, & r < 0,\end{cases} \qquad H_b = (\mu_{i+j-3})_{0\leq i,j<b},$$

$$R_b = \operatorname{diag}_{0\leq i<b}(2b+2i-7)!, \qquad E_b = \operatorname{diag}_{0\leq k<b}\frac{1}{(2b-2k)!}.$$

For every integer $b \geq 8$ the inherited reduction is

$$\frac{\det M_b}{\det R_b \det E_b \det H_b} = \frac{P(b; 2B_{2b-2}, \dots, 2B_{2b-14})}{576(2b-5)D_8(b)}. \tag{2.9}$$

The matrix H_b is nonsingular. One short proof conjugates it by $\operatorname{diag}(1,-1,1,-1,\dots)$ and partitions after the first two indices. The lower-right block is a positive definite moment matrix on the points $(2\pi m)^{-2}$, $m \geq 1$. The upper-left block is zero, and the rectangular off-diagonal block has row rank two: its first two columns form a nonsingular antitriangular matrix. Its Schur complement is negative definite. Hence H_b has exactly two negative eigenvalues and is nonsingular. Thus nonvanishing of the numerator of (2.9) proves the rank assertion.

All formulas in this subsection are inherited definitions or reductions. The proof below supplies the previously missing global nonvanishing. In particular, the symbolic certificate is used as a finite algebraic identity, not as an extrapolation from values of b.

2.3 The highest-degree part is a zeta moment determinant

Write $P = P_4 + P_{\leq 3}$ according to total degree in $U_1, \dots, U_7$. For an integer $b \geq 8$, set

$$F_0 = (2b-14)!, \qquad \alpha_j = \frac{(2b-2j)!}{F_0}, \qquad V_j = \frac{U_j}{\alpha_j} \quad (1 \leq j \leq 7),$$

and define

$$\begin{aligned}G(b) = 12288&(b-6)^4(b-5)^4(b-4)^4(b-3)^3(b-2)^2(b-1)\\ &\cdot (2b-13)^4(2b-11)^4(2b-9)^4(2b-7)^3(2b-5)^3(2b-3)^2(2b-1).\end{aligned} \tag{2.10}$$

Lemma 2.2 (Highest-degree Hankel factorization)**.** *The exact identity is*

$$P_4(b; U) = G(b)\det(V_{i+j+1})_{0\leq i,j\leq 3}. \tag{2.11}$$

At the actual doubled Bernoulli values, this becomes

$$P_4(b; 2B_{2b-2}, \dots, 2B_{2b-14}) = \frac{256G(b)F_0^4}{(2\pi)^{8b-32}}\det\big(\zeta(2b-14+2i+2j)\big)_{0\leq i,j\leq 3} > 0. \tag{2.12}$$

Proof. This factorization has a direct structural proof. In (2.4), substitute $z = N_{i\ell} - y$. Then

$$W_{i\ell}(N_{i\ell} - y) = (2n_i + 2 - y)(4n_i + 2 - y)^{\underline{2i}},$$

which is independent of ℓ. Write its Newton coefficients as d_{ih}. In particular,

$$d_{i1} = -(i+1)(4n_i+2)^{\underline{2i}}. \tag{2.13}$$

Indeed the coefficient is the value at $y=1$ minus the value at $y=0$; using $(x-1)^{\underline{2i}} = x^{\underline{2i}}(x-2i)/x$ and $4n_i+2 = 2(2n_i+1)$ gives (2.13).

Let $T = \mathcal{A} - 2F$. In a term with $h = 2k+1$, the Bernoulli variable index is $j = 4-i+\ell+k$ and $N_{i\ell} - h = 2b-2j$. Thus substitution of $U_j = \alpha_j V_j$ cancels the factorials in (2.5). Define

$$\rho_i = \frac{1}{(2n_i+1)(2n_i+2)}, \qquad \sigma_\ell = \frac{R_\ell!}{F_0(2\ell+2)!},$$

and the lower triangular matrix L by $L_{i,i-k} = d_{i,2k+1}$ for $0 \le k \le i$. Then the exact matrix identity is

$$T = \operatorname{diag}(\rho_i)\, L\, (V_{4-i+\ell})_{0\le i,\ell\le 3} \operatorname{diag}(\sigma_\ell).$$

Reversing its four middle rows has sign $(-1)^6 = 1$ and gives the Hankel matrix in (2.11). The diagonal entries of L are (2.13). Taking determinants, multiplying by $576(2b-5)D_8(b)/\det F(b)$, and cancelling the finite products in (2.8) gives exactly (2.10). Since the highest-degree part of $\det(2F+T)$ is $\det T$, this proves (2.11). An independent coefficient check in the accompanying script verifies all sixteen Hankel monomials.

Euler's Bernoulli formula [13, Eq. 24.8.1] gives

$$V_j = (-1)^{b-j+1} \frac{4F_0}{(2\pi)^{2b-2j}} \zeta(2b-2j).$$

Factoring the row and column signs and powers of 2π in the determinant, and then reversing both its rows and its columns, gives (2.12). All signs cancel. Its positivity follows from the strictly positive moment representation proved next. □

Lemma 2.3 (An explicit positive contribution)**.** *For every real $n > 1/2$,*

$$\det\big(\zeta(2n+2i+2j)\big)_{0\le i,j\le 3} \ge \frac{30625}{48922361856}\, 24^{-2n} > 0. \tag{2.14}$$

Proof. With $x_m = m^{-2}$, the entries are the moments $\sum_{m\ge 1} x_m^{n+i+j}$. Cauchy–Binet for a finite truncation, followed by convergence of each entry, yields

$$\det\big(\zeta(2n+2i+2j)\big) = \sum_{m_1<m_2<m_3<m_4} (m_1m_2m_3m_4)^{-2n} \prod_{i<j} (m_i^{-2} - m_j^{-2})^2.$$

Every summand is positive. Keeping the term $(1,2,3,4)$ proves (2.14), since

$$\prod_{1\le i<j\le 4} (i^{-2} - j^{-2})^2 = \frac{30625}{48922361856} =: V.$$

No limiting determinant sign is inferred numerically. □

2.4 Uniform domination of every lower-degree term

The integer polynomial defined by (2.7) has 94 monomials when grouped by the seven U variables. Every coefficient of $P_{\le 3}$ has degree at most 15 in b, and its full integer coefficient norm is exactly

$$K := \sum_{\substack{\text{monomials } cb^rU^\nu \text{ of } P \\ |\nu|\le 3}} |c| = 1465775272343372. \tag{2.15}$$

This finite integer is obtained by exact coefficient extraction, independently replayed from the source polynomial certificate. It is an upper bound on the whole lower-degree contribution, with no cancellation assumed.

For $b \geq 512$, let

$$T_b = \frac{8(2b)^{12}(2b-14)!}{(2\pi)^{2b-14}}.$$

Every actual variable satisfies $|2B_{2b-2j}| \leq T_b$: use $\zeta(2b-2j) < 2$, bound the factorial ratio by $(2b)^{12}$, and use $2\pi > 1$ for the remaining nonnegative exponent. Also $T_b \geq 1$. For example $20! > 8^{20}$, the ratio for subsequent even factorials is greater than 8^2, and $2\pi < 8$. Consequently

$$|P_{\leq 3}(b; 2B_{2b-2}, \ldots, 2B_{2b-14})| \leq Kb^{15}T_b^3. \tag{2.16}$$

For $b \geq 14$, every $b-j$ factor in (2.10) is at least $b/2$, and every $2b-j$ factor is at least b. There are 18 factors of the first kind and 21 of the second, with multiplicity. Thus

$$G(b) \geq \frac{3}{64}b^{39}.$$

Combining this with (2.12) and (2.14) gives

$$P_4 \geq \frac{12Vb^{39}((2b-14)!)^4}{(2\pi)^{8b-32}}\,24^{-2b+14}.$$

Dividing (2.16) by this expression, first cancelling the common powers of 2π and then using $2\pi < 8$, proves

$$\frac{|P_{\leq 3}|}{P_4} \leq R_b := C\,\frac{b^{12}192^{2b}}{(2b-14)!}, \qquad C = \frac{K2^{73}}{3V24^{14}} = \frac{2112404790507039177735799664803 84}{602791875}. \tag{2.17}$$

The cancellation before replacing 2π is useful: it avoids artificially weakening the exponential factor.

All that remains here is elementary exact integer arithmetic:

$$R_{512} < 2^{-675} < 1, \qquad \frac{R_{b+1}}{R_b} = \left(1+\frac{1}{b}\right)^{12}\frac{192^2}{(2b-12)(2b-13)} < 1 \quad (b \geq 512). \tag{2.18}$$

The first inequality means the exact integer comparison

$$2^{675}\,211240479050703917773579966480384\,512^{12}192^{1024} < 602791875 \cdot 1010!,$$

which is verified in the accompanying script. For the second, $(513/512)^{12} < 2$ and $2 \cdot 192^2 < 1011 \cdot 1012$ suffice. It follows that $P = P_4 + P_{\leq 3} > 0$ for every $b \geq 512$. The threshold is chosen for a transparent uniform proof; no minimal threshold is asserted.

2.5 Finite completion of the omitted residue class

For completeness, the inherited congruence proof gives, for $b \geq 11$,

$$P(b; 2B_{2b-2}, \ldots, 2B_{2b-14}) \equiv 4(b-1)(4b+15) \pmod{64} \quad \text{in } \mathbb{Z}_{(2)}.$$

It uses $2B_{2m} \equiv 1 + 4m + 16m^2 \pmod{64}$ for $m \geq 4$, together with its symbolic determinant certificate. Thus $P \neq 0$ when $b \not\equiv 1 \pmod{16}$. The source checks $5 \leq b \leq 10$ directly. Its congruence assertion does not supply the omitted residue class, and no extrapolation of its valuation formula is used here.

Below 512 the only remaining indices are $b = 16k+1$, $1 \leq k \leq 31$. The following table records the exact valuation of P at every such index.

b	$v_2(P)$	b	$v_2(P)$	b	$v_2(P)$	b	$v_2(P)$
17	6	33	7	49	6	65	10
81	6	97	7	113	6	129	8
145	6	161	7	177	6	193	9
209	6	225	7	241	6	257	8
273	6	289	7	305	6	321	11
337	6	353	7	369	6	385	8
401	6	417	7	433	6	449	9
465	6	481	7	497	6		

Every entry is certified by a nonzero residue modulo 2^{32}; all 31 residues are included in the machine-readable record. They can be obtained either from exact rational Bernoulli numbers or entirely in modular arithmetic using

$$(2m+1)\,(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1} \binom{2m+1}{2j}\,(2B_{2j}).$$

The divisor $2m+1$ is odd, so the recurrence is valid modulo 2^{32}. Two independent implementations agree on every residue. In particular no finite numerical approximation is treated as a nonvanishing certificate. As an explicit first new value,

$$\frac{\det M_{17}}{\det R_{17}\det E_{17}\det H_{17}} = -\frac{5564518155020169678178976040991}{8937041637533625}.$$

This settles the previously omitted pair $(p,q) = (8,34)$. The inherited cases, this finite table, and (2.18) cover every $b \geq 5$, proving Theorem 2.1.

2.6 An all-order family of ordinary polygamma identities

The rank theorem supplies explicit identities as well as a decision about a matrix. For $b \geq 5$ and $0 \leq k < b$, define

$$w_{k,r}^{(b)} = \sum_{t=0}^{b-1} (M_b^{-1})_{kt} \binom{2(b+t)+1}{2r} 2^{-2(b+t-r)}, \tag{2.19}$$

$$Q_{b,k}(x) = \sum_{r=0}^{2b-1} w_{k,r}^{(b)} x^{2r}, \tag{2.20}$$

where the summand is zero for $r > b+t$. These are finite rational formulas for every coefficient. Put

$$\eta_{s,0} = \frac{1}{s-1}, \qquad \eta_{s,j} = \frac{(s)_{2j-1} B_{2j}(1/2)}{(2j)!} \quad (j \geq 1), \qquad c_{b,\ell} = \sum_{j=0}^{\ell} \eta_{8,j}\eta_{2b,\ell-j}.$$

Let $S_{b,k}(x)$ be the polynomial formed by retaining the nonnegative powers in the formal product

$$Q_{b,k}(x)x^{-2b-6}\sum_{\ell\geq 0} c_{b,\ell}x^{-2\ell}.$$

Only finitely many terms enter this polynomial, and its degree is at most $2b-8$. Then

$$\boxed{\sum_{n=0}^{\infty}\left\{Q_{b,k}(x_n)\frac{\psi^{(7)}(n+1)\psi^{(2b-1)}(n+1)}{7!\,(2b-1)!} - S_{b,k}(x_n)\right\} = \zeta(2k).} \tag{2.21}$$

This is an ordinary absolutely convergent series. Indeed (2.19) and (2.1) identify its continued moment with the kth row of $M_b^{-1}M_b$, giving $\zeta(2k)$. The centered Hurwitz expansion is

$$A_8(n)A_{2b}(n) \sim x_n^{-2b-6} \sum_{\ell \geq 0} c_{b,\ell} x_n^{-2\ell}.$$

The polynomial $S_{b,k}$ removes every nondecaying term, leaving $O(x_n^{-2})$. Each removed nonnegative even power has centered continued sum $\zeta(-2j, 1/2) = 0$, so the ordinary convergent sum has precisely the same value. Thus no unspecified summation of a divergent series remains in (2.21).

To reconstruct a later moment, first use (2.1) on the initial prefix, invert M_b, form every later C_j from the source Bernoulli-binomial formula, and finally use (2.2). Every operation is finite rational arithmetic. This completes the order-eight family. The still-uncovered pairs have smaller even order at least ten and order gap at least six; their optimal-prefix conjecture remains separate.

3 Effective eventual reconstruction for every fixed even order

The structural factorization used for order eight has an all-order consequence. It reduces the full conjecture at each fixed smaller order to an explicitly bounded finite calculation. It does not prove that all those finitely many matrices are nonsingular.

Theorem 3.1 (Effective eventual optimal-prefix rank)**.** *For every integer $a \geq 1$ there is a constructible integer B_a such that the optimal-prefix coefficient matrix $M_{a,b}$ is nonsingular for every integer $b \geq B_a$. Thus, for every fixed even tail order $p = 2a$, only finitely many larger even orders can possibly fail the source optimal-prefix conjecture. The formulas below give a terminating algorithm for an integer B_a, using finite polynomial arithmetic and exact integer comparisons.*

Here and below, the general source matrix is

$$(M_{a,b})_{ik} = B_{2(i+k-a+1)} \left\{ \frac{\binom{2i+2}{2a-2k}}{2i+2} + \frac{\binom{2b+2i-2a+2}{2b-2k}}{2b+2i-2a+2} \right\}, \quad 0 \leq i,k < b. \tag{3.1}$$

Negative Bernoulli indices give zero; each binomial summand is omitted unless its lower index lies between 2 and its upper index. This is the same normalization as (2.3) at $a = 4$.

3.1 Exact normalization of the retained determinant

Assume initially $b \geq 2a$. For $0 \leq i, \ell < a$ define

$$n_i = b - a + i, \quad N_{i\ell} = 2(n_i - \ell) + 1, \quad R_\ell = 2b - 2a + 2 + 2\ell,$$

$$F_{i\ell} = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n_i - r)(2n_i + 3 + r),$$

and define integer-polynomial coefficients d_{ih} by

$$(2n_i + 2 - y)(4n_i + 2 - y)^{\underline{2i}} = \sum_{h=0}^{2i+1} d_{ih} y^{\underline{h}}. \tag{3.2}$$

They lie in $\mathbb{Z}[b]$: the change of basis from ordinary powers to falling powers has integer Stirling-number coefficients. Let $U_1, \ldots, U_{2a-1}$ be independent variables and put

$$\mathcal{A}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)!N_{i\ell}!} \sum_{k=0}^{i} d_{i,2k+1} N_{i\ell}^{\underline{2k+1}} U_{a-i+\ell+k}. \tag{3.3}$$

The inherited reduction is the theorem "Reduction to a determinant of size a," equation `eq:general-small-det`, in the source file `03b_general_hankel.tex` [3]. We give its exact multiplier and the normalization argument, so that no nonvanishing statement about an uncomputed inverse matrix is needed.

Lemma 3.2 (Exact source-to-retained determinant multiplier)**.** *Put*

$$h_n = \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!}, \qquad c_n = \frac{(2n+1)(2n+2)!}{(4n+2)!}, \qquad \epsilon_{a,b} = a(b-a) + \frac{a(a-1)}{2}.$$

For $b \geq 2a$, with $U_j = 2B_{2b-2j}$,

$$\det M_{a,b} = (-1)^{\epsilon_{a,b}} \frac{\prod_{t=0}^{b-1}(2b+2t-2a+1)!}{\prod_{k=0}^{b-1}(2b-2k)!} \left(\prod_{n=0}^{b-a-1} h_n\right)\left(\prod_{i=0}^{a-1} c_{n_i}\right) \det \mathcal{A}. \tag{3.4}$$

Every scalar factor other than the displayed sign is strictly positive. In particular,

$$\det M_{a,b} \neq 0 \quad \Longleftrightarrow \quad \det \mathcal{A}(b; 2B_{2b-2}, \ldots, 2B_{2b-4a+2}) \neq 0. \tag{3.5}$$

Proof. Set $\mu_j = B_{2j}/(2j)!$ for $j \geq 0$ and $\mu_j = 0$ for $j < 0$. Cancelling factorials in (3.1) gives

$$M_{a,b} = R(H + U^{\mathrm{pert}})E, \quad R_{tt} = (2b+2t-2a+1)!, \quad E_{kk} = \frac{1}{(2b-2k)!},$$

where $H_{tk} = \mu_{t+k-a+1}$ and

$$U^{\mathrm{pert}}_{tk} = \mathbf{1}_{k<a}\mu_{t+k-a+1}\frac{(2t+1)!(2b-2k)!}{(2a-2k)!(2b+2t-2a+1)!}.$$

The diagonal symbols R, E in this proof are matrices; R_ℓ in (3.3) is the previously defined integer.

The explicit monic Padé rows are

$$\pi_n(x) = \sum_{t=0}^{n} p_{nt}x^t, \qquad p_{nt} = \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!}.$$

For $\mathcal{N}(x^r) = \mu_{r+1}$ they satisfy

$$\mathcal{N}(x^j\pi_n) = 0 \ (0 \leq j < n), \qquad \mathcal{N}(x^n\pi_n) = h_n. \tag{3.6}$$

Here is a finite verification of the normalization. Define

$$P_N(t) = \sum_{k=0}^{N} \frac{(2N-k)!N!}{(2N)!k!(N-k)!}t^k.$$

Direct coefficient comparison gives

$$e^t P_N(-t) - P_N(t) = \frac{(-1)^N(N!)^2}{(2N)!(2N+1)!}t^{2N+1} + O(t^{2N+2}).$$

Indeed, the coefficients through $2N$ reduce to the finite identity $\sum_k (-1)^k\binom{N}{k}(1+x)^{2N-k} = x^N(1+x)^N$; the next coefficient is $(2N)!^{-1}\int_0^1 x^N(x-1)^N\,dx$. Taking $N = 2n+1$, set

$$q_n(t^2) = \frac{P_N(t) - P_N(-t)}{t}, \qquad r_n(t^2) = \frac{P_N(t) + P_N(-t)}{2}.$$

For $\Phi(z) = \sum_{j\geq 0}\mu_j z^j = \frac{\sqrt{z}}{2}\coth\frac{\sqrt{z}}{2}$ this proves

$$\Phi(z)q_n(z) - r_n(z) = h_n z^{2n+1} + O(z^{2n+2}), \qquad \pi_n(x) = x^n q_n(1/x).$$

Coefficient comparison proves (3.6); it also gives, for $n \ge \ell \ge 0$,

$$\sum_{t=0}^{n} p_{nt}\mu_{t-\ell} = c_n \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n-r)(2n+3+r). \tag{3.7}$$

Specifically, this sum is $[z^{n-\ell}]r_n(z)$, and its factorial coefficient formula is the right side. Thus both c_n and h_n have been obtained without an assumed Padé normality assertion.

Replace row n of $H+U^{\mathrm{pert}}$ by the linear combination with coefficients p_{nt}, for $0 \le n < b$. This simultaneous transformation is unit lower triangular. In columns $k \ge a$ the result is $\mathcal{N}(x^{k-a}\pi_n)$. Hence the final a rows vanish there, while the first $b-a$ rows in those columns form an upper triangular square matrix with diagonal $h_0, \dots, h_{b-a-1}$. The retained a-square matrix is

$$A^{\mathrm{ret}}_{ik} = \sum_{t=0}^{n_i} p_{n_i t}\mu_{t+k-a+1} \left(1 + \frac{(2t+1)!(2b-2k)!}{(2a-2k)!(2b+2t-2a+1)!}\right).$$

Moving the first a columns past the others contributes $(-1)^{a(b-a)}$. Reverse the retained columns, putting $\ell = a-1-k$, and divide row i by c_{n_i}. This adds the determinant factor $(-1)^{a(a-1)/2}\prod_i c_{n_i}$.

It remains to identify this divided matrix with (3.3). Its unperturbed entry is $F_{i\ell}$ by (3.7). Writing $n = n_i$, $N = N_{i\ell}$, and

$$W(j) = (j+2\ell+1)(2n+2\ell+j+1)^{\underline{2i}}, \qquad C = \frac{R_\ell!}{(2n+1)(2n+2)(2\ell+2)!N!},$$

factorial cancellation gives the perturbation entry $2C\sum_{r=0}^{n-\ell}\binom{N}{2r}B_{2r}W(2r)$. Since $W(N-y)$ is exactly the polynomial in (3.2), use the finite identity

$$\sum_{j=0}^{N} \binom{N}{j} B_j (N-j)^{\underline{h}} = N^{\underline{h}} B_{N-h}(1).$$

It follows immediately by taking $N^{\underline{h}}$ outside the binomial sum and using the defining translation formula for Bernoulli polynomials. Here $N \ge 2i+3$, so all indices $N-h$ are at least two. Only odd h survive. Restoring the omitted $j=1$ term, with $B_1 = -1/2$, gives

$$\sum_{r} \binom{N}{2r} B_{2r}W(2r) = \sum_{k=0}^{i} d_{i,2k+1} N^{\underline{2k+1}} B_{2b-2(a-i+\ell+k)} + \frac{N}{2}W(1).$$

Finally, $CNW(1) = F_{i\ell}$ by factorial cancellation. The divided retained entry is therefore $2F_{i\ell}$ plus the Bernoulli term in (3.3). Restoring all row and column factors proves (3.4). □

For reference, the unperturbed base determinant has the exact normalization

$$\det H = (-1)^{\epsilon_{a,b}} \left(\prod_{n=0}^{b-a-1} h_n\right) \left(\prod_{i=0}^{a-1} c_{n_i}\right) \det F \neq 0. \tag{3.8}$$

One can verify the last inequality without using it in the lemma. Conjugate H by $\mathrm{diag}(1,-1,\dots)$. Euler's Bernoulli formula shows that entries with $t+k \ge a$ come from a strictly definite positive moment matrix, with sign $(-1)^a$. For $a = 2h$ the resulting block matrix is $\left(\begin{smallmatrix} 0 & B \\ B^{\mathsf{T}} & P \end{smallmatrix}\right)$, where P is positive definite and B has full row rank h. For $a = 2h+1$ it is $\left(\begin{smallmatrix} ee^{\mathsf{T}} & B \\ B^{\mathsf{T}} & -P \end{smallmatrix}\right)$, where P is positive definite, $e = (0,\dots,0,1)^{\mathsf{T}}$, and the first h rows of B have a nonsingular antitriangular minor. The Schur complements are respectively strictly negative and strictly positive. The inertia is $(b-h,h)$ in the even case and $(h+1,b-h-1)$ in the odd case. Together with (3.8), this also proves $\det F > 0$. Strict positivity of the moment blocks follows from $\sum_{m\ge1}(2\pi m)^{-2s}P_0((2\pi m)^{-2})^2 > 0$ for a nonzero polynomial P_0 and $s \ge 1$.

3.2 The general highest-degree determinant identity

All denominators can be cleared constructively. Set

$$J_a(b)=\prod_{i=0}^{a-1}(2n_i+1)(2n_i+2),$$
$$E_a(b)=\prod_{i,\ell=0}^{a-1}\ \prod_{r=1}^{\max\{0,2i-4\ell-1\}}(R_\ell+r),$$
$$D_a(b)=(2a)!J_a(b)E_a(b),\qquad e_a=\sum_{i,\ell=0}^{a-1}\max\{0,2i-4\ell-1\}.$$

The symbols $E_a(b)$ and $D_a(b)$ here are auxiliary polynomials; they are unrelated to the moment coordinates. Because $N_{i\ell}-R_\ell=2i-4\ell-1$, the polynomial D_a clears every negative factorial ratio in (3.3), every row denominator, and every constant factorial denominator. Consequently

$$\widetilde{\mathcal{A}}=D_a\mathcal{A}\in\mathbb{Z}[b,U_1,\dots,U_{2a-1}]^{a\times a},\qquad \mathscr{P}_a=\det\widetilde{\mathcal{A}}\in\mathbb{Z}[b,U_1,\dots,U_{2a-1}].$$

Also $D_a(b)>0$ for $b\ge 2a$.

Put

$$F_0=(2b-4a+2)!,\qquad V_j=U_j\frac{F_0}{(2b-2j)!},\qquad \kappa_a=\frac{a!((2a)!)^a}{\prod_{\ell=0}^{a-1}(2\ell+2)!},$$

and

$$\Gamma_a(b)=\kappa_aJ_a(b)^{a-1}E_a(b)^a\prod_{i=0}^{a-1}(4n_i+2)^{\underline{2i}}\prod_{\ell=0}^{a-1}\prod_{r=1}^{2a+2\ell}(2b-4a+2+r).\tag{3.9}$$

The subscript $[a]$ denotes the part homogeneous of total degree a in the U variables. The exact factorization is

$$(\mathscr{P}_a)_{[a]}=(-1)^{a(a+1)/2}\Gamma_a(b)\det(V_{i+j+1})_{0\le i,j<a}.\tag{3.10}$$

To prove it, write $T=\mathcal{A}-2F$ and substitute the V variables. For the term $U_{a-i+\ell+k}$, $N_{i\ell}-(2k+1)=2b-2(a-i+\ell+k)$, so all nonconstant factorials cancel. Exactly as in the order-eight proof,

$$T=\operatorname{diag}\!\left(\frac{1}{(2n_i+1)(2n_i+2)}\right)L\,(V_{a-i+\ell})_{0\le i,\ell<a}\operatorname{diag}\!\left(\frac{R_\ell!}{F_0(2\ell+2)!}\right),$$

where $L_{i,i-k}=d_{i,2k+1}$ and $d_{i1}=-(i+1)(4n_i+2)^{\underline{2i}}$. The a diagonal signs of L, together with row reversal of the middle matrix, give $(-1)^{a(a+1)/2}$. The remaining products are precisely (3.9) after multiplication by D_a^a. This proves (3.10) for every a, without evaluating any large determinant experimentally.

At $U_j=2B_{2b-2j}$, Euler's Bernoulli formula gives

$$\left|(\mathscr{P}_a)_{[a]}\right|=\frac{\Gamma_a(b)(4F_0)^a}{(2\pi)^{2a(b-a)}}\det\bigl(\zeta(2b-4a+2+2i+2j)\bigr)_{0\le i,j<a}.\tag{3.11}$$

Its sign is $(-1)^{ab+a(a+1)/2}$. Define the positive rational number

$$V_a=\prod_{1\le r<s\le a}(r^{-2}-s^{-2})^2,\qquad V_1=1.$$

Keeping the first a support points in Cauchy–Binet proves

$$\det\bigl(\zeta(2b-4a+2+2i+2j)\bigr)\ge V_a(a!)^{-2b+4a-2}>0.\tag{3.12}$$

Thus the highest-degree term is always nonzero in the stated range.

3.3 An exact algorithm for a sufficient threshold

Let $\|f\|_1$ denote the sum of the absolute values of all integer coefficients of a polynomial in $b, U_1, \dots, U_{2a-1}$. Compute

$$E_a^* = \max_{0\le i,\ell<a} \|\widetilde{\mathcal{A}}_{i\ell}\|_1, \qquad K_a = a!(E_a^*)^a, \qquad g_a = 6a^2 - 4a + ae_a.$$

These are finite integer calculations on only a^2 explicitly given entries; no expansion of the whole determinant is required. The coefficient norm is submultiplicative, so the full coefficient norm of $\mathscr{P}_a$ is at most K_a.

Each scaled entry has degree at most $6a + e_a - 4$ in b. For the nonconstant part this follows from $\deg_b d_{ih} \le 2i + 1 - h$, the degree h of $N_{i\ell}^{\underline{h}}$, and the degree difference $R_\ell - N_{i\ell} = 1 + 4\ell - 2i$; the total bound is $\deg D_a + 4\ell$. The same bound holds for $D_a F_{i\ell}$. Hence $\deg_b \mathscr{P}_a \le g_a$. The factored polynomial (3.9) also has degree exactly g_a. Every linear factor in it is at least b when $b \ge 4a$, so

$$\Gamma_a(b) \ge \kappa_a b^{g_a} \quad (b \ge 4a).$$

For $b \ge 4a + 10$, define

$$T_{a,b} = \frac{8(2b)^{4a-4}F_0}{(2\pi)^{2b-4a+2}} \ge 1.$$

For the last inequality, put $s_0 = 2b - 4a + 2 \ge 26$; $26! > 8^{26}$ implies $s_0! > 8^{s_0}$ by induction, and $2\pi < 8$. As before, $|2B_{2b-2j}| \le T_{a,b}$ for $1 \le j \le 2a - 1$. Since the remaining terms have total U degree at most $a - 1$,

$$|(\mathscr{P}_a)_{[\le a-1]}| \le K_a b^{g_a} T_{a,b}^{a-1}.$$

Combining this with (3.11) and (3.12), cancelling the common powers of 2π, and then using $1 < 2\pi < 8$, gives

$$\frac{|(\mathscr{P}_a)_{[\le a-1]}|}{|(\mathscr{P}_a)_{[a]}|} \le R_a(b) := C_a \frac{b^{L_a}(8a!)^{2b}}{(2b - 4a + 2)!}, \tag{3.13}$$

where every constant is explicit and rational:

$$\begin{aligned} L_a &= 4(a-1)^2, \qquad \chi_a = 2a^2 - 6a + 2, \\ C_a &= \frac{K_a\, 8^{a-1+\max(\chi_a,0)} 2^{L_a}}{\kappa_a 4^a V_a (a!)^{4a-2}}. \end{aligned} \tag{3.14}$$

The use of $8^{\max(\chi_a,0)}$ also covers $a = 1, 2$, where χ_a is negative.

Here is a terminating purely arithmetic prescription. Start at

$$B_a^{(0)} = \max\{4a + 10,\ L_a,\ 16a!\},$$

and increase the integer b until the exact rational comparison $R_a(b) < 1$ holds. Define B_a to be that first integer. The procedure must terminate, because for every $b \ge B_a^{(0)}$,

$$\frac{R_a(b+1)}{R_a(b)} = \left(1 + \frac{1}{b}\right)^{L_a} \frac{(8a!)^2}{(2b - 4a + 4)(2b - 4a + 3)} < \frac{3}{4}.$$

Indeed $b \ge L_a$ gives $(1 + 1/b)^{L_a} < 3$, $b \ge 4a$ makes both denominator factors at least b, and $b \ge 16a!$ supplies the last numerical factor. The comparison uses integer factorials and powers after clearing the fixed rational denominator of C_a.

For every $b \ge B_a$, (3.13) is strictly less than one, so the highest-degree term cannot be cancelled. Equations (3.5) and $D_a(b) > 0$ prove Theorem 3.1. In particular, for each input a one may inspect the finite set $a < b < B_a$ exactly: this either proves the entire fixed-order family or produces a genuine counterexample. The construction does not perform those finite calculations for all a, and does not settle the unrestricted conjecture.

a	Smaller order $p = 2a$	Sufficient B_a	Guaranteed even q
1	2	16	$q \geq 32$
2	4	45	$q \geq 90$
3	6	142	$q \geq 284$
4	8	467	$q \geq 934$
5	10	1920	$q \geq 3840$

Table 2: Exact sufficient thresholds returned by the constructive algorithm. No minimality is asserted.

The first four rows of Table 2 are exact diagnostics of the general construction; complete-family results already establish reconstruction for every larger even order in those cases, including the order-eight theorem proved here. The fifth row gives a concrete further consequence: for smaller order $p = 10$, optimal-prefix reconstruction holds for every even $q \geq 3840$. The algorithm computes $B_5 = 1920$ by an exact rational comparison. It does not check the remaining finite range $6 \leq b < 1920$.

4 All regular centered quartic trigamma moments

The incoming reconstruction report asks for an explicit quartic extension of its centered cubic moments, with a fixed coordinate space and a generator [2, Question 5 and Section 6]. We answer that question. The starting, unweighted quartic sum is already evaluated explicitly by Hoffman [9, Corollary 4.4 and the following example]. The contribution here is the complete family of centered polynomial-subtracted moments, the finite Bernoulli formula connecting it to the cubic family, and the generator with all initial constants fixed.

4.1 Definition and ordinary summation convention

Throughout this section set

$$A_n = \psi'(n+1) = \zeta(2) - H_n^{(2)}, \qquad x_n = n + \tfrac{1}{2}, \qquad \mathcal{V}(s) = \sum_{n=0}^{\infty} A_n^4 x_n^{-s} \quad (\Re s > -3). \tag{4.1}$$

Use $B_1 = -1/2$ for ordinary Bernoulli numbers. Define

$$h_k = \sum_{a+b+c+d=k} B_{2a}(1/2)B_{2b}(1/2)B_{2c}(1/2)B_{2d}(1/2), \qquad a, b, c, d \geq 0. \tag{4.2}$$

For example

$$h_0 = 1, \qquad h_1 = -\frac{1}{3}, \qquad h_2 = \frac{19}{120}, \qquad h_3 = -\frac{1871}{15120}.$$

Proposition 4.1 (Regular parity and prescribed subtractions)**.** *The only poles of $\mathcal{V}$ are simple poles at $s = -3 - 2k$, $k \geq 0$, with residues h_k. In particular all the values*

$$V_r = \mathcal{V}(-2r), \qquad r \geq 0, \tag{4.3}$$

are regular. They are represented by the ordinary absolutely convergent series

$$\boxed{V_r = \sum_{n=0}^{\infty} \left(x_n^{2r} A_n^4 - \sum_{k=0}^{r-2} h_k x_n^{2r-4-2k} \right).} \tag{4.4}$$

The inner sum is empty for $r = 0, 1$.

Proof. The centered Hurwitz asymptotic expansion gives

$$A_n \sim \sum_{j\ge 0} B_{2j}(1/2)x_n^{-2j-1}, \qquad A_n^4 \sim \sum_{k\ge 0} h_k x_n^{-4-2k}.$$

Subtract an arbitrary finite number of terms and add the corresponding Hurwitz functions $h_k\zeta(s+4+2k,1/2)$. The remainder series is normally convergent in successively larger left half-planes. This continues $\mathcal{V}$ meromorphically and gives the listed poles and residues. Every term in the definition of h_k has sign $(-1)^k$, because $B_{2j}(1/2)$ has sign $(-1)^j$, including $j=0$. Consequently $h_k \neq 0$ and the poles are actual.

At $s=-2r$ the added terms corresponding to $0\le k\le r-2$ are $h_k\zeta(4+2k-2r,1/2)=0$, since $\zeta(-2m,1/2)=0$ for every integer $m\ge 0$. The remaining summand in (4.4) is $O(x_n^{-2})$; for $r=0$ it is even $O(x_n^{-4})$. This proves the asserted ordinary identity. □

4.2 The finite formula

We use two explicitly evaluated starting constants:

$$Q_5 = 4\zeta(2)\zeta(3) - 10\zeta(5), \tag{4.5}$$

$$Q_7 = 51\zeta(3)\zeta(4) + 10\zeta(2)\zeta(5) - \frac{301}{4}\zeta(7). \tag{4.6}$$

The cubic coefficients inherited from the incoming report are

$$C_1 = 9\zeta(3) - \frac{17}{4}\zeta(4), \tag{4.7}$$

$$C_j = 9B_{2j-2}\zeta(3) - \frac{1}{2}\delta_{j,2}\zeta(2) + R_j, \qquad j\ge 2, \tag{4.8}$$

$$R_j = -\frac{3}{2(2j-1)}\sum_{h=0}^{j-2}(2h+1)\binom{2j-1}{2h+3}B_{2j-4-2h}B_{2h} + \frac{1}{4}\delta_{j,3}. \tag{4.9}$$

These formulas are reproduced to make the new quartic answer directly computable; their cubic evaluation is not a new claim of this section.

Define $D_0 = Q_7$ and, for $j\ge 1$, set

$$\boxed{D_j = \frac{2}{j}\sum_{\ell=1}^{j}\binom{2j}{2\ell}B_{2j-2\ell}C_\ell + \varepsilon_j, \qquad \varepsilon_j = \begin{cases} Q_5, & j=1,\\ -3\zeta(3), & j=2,\\ \dfrac{2j-5}{2}B_{2j-6}, & j\ge 3.\end{cases}} \tag{4.10}$$

In particular

$$\begin{aligned} D_1 &= 2C_1 + Q_5,\\ D_2 &= \frac{15}{2}\zeta(3) - \frac{17}{4}\zeta(4) - \frac{1}{2}\zeta(2) - \frac{1}{2},\\ D_3 &= -\frac{7}{10}\zeta(3) + \frac{17}{12}\zeta(4) - \frac{5}{6}\zeta(2) - \frac{3}{5},\\ D_4 &= \frac{23}{35}\zeta(3) - \frac{17}{12}\zeta(4) + \frac{7}{12}\zeta(2) - \frac{81}{560}. \end{aligned} \tag{4.11}$$

Theorem 4.2 (Complete quartic Bernoulli formula)**.** *For every $r\ge 0$,*

$$\boxed{V_r = \frac{1}{2r+1}\sum_{j=0}^{r}\binom{2r+1}{2j+1}B_{2r-2j}(1/2)D_j.} \tag{4.12}$$

Consequently every sum (4.4) lies in the fixed space

$$\operatorname{span}_{\mathbb{Q}}\{Q_7, Q_5, \zeta(2), \zeta(3), \zeta(4), 1\}. \tag{4.13}$$

The first six moments span all six formal coordinates: in the order displayed in (4.13), their coefficient determinant is

$$\det[V_0, \dots, V_5]_{\text{coordinates}} = \frac{7361}{49896000} \neq 0. \tag{4.14}$$

This is a statement about the specified coefficient formulas, not an assertion of arithmetic independence of these six real constants.

Here are the first four identities in compact form:

$$V_0 = Q_7, \tag{4.15}$$

$$V_1 = \frac{Q_5}{3} - \frac{Q_7}{12} + 6\zeta(3) - \frac{17}{6}\zeta(4), \tag{4.16}$$

$$V_2 = -\frac{Q_5}{6} + \frac{7Q_7}{240} - \frac{1}{10}\zeta(2) - \frac{3}{2}\zeta(3) + \frac{17}{30}\zeta(4) - \frac{1}{10}, \tag{4.17}$$

$$V_3 = \frac{7Q_5}{48} - \frac{31Q_7}{1344} + \frac{1}{168}\zeta(2) + \frac{13}{20}\zeta(3) + \frac{17}{672}\zeta(4) + \frac{11}{280}. \tag{4.18}$$

The first two are convergent without subtractions. The next two are, respectively,

$$\sum_{n\geq 0}(x_n^4 A_n^4 - 1), \qquad \sum_{n\geq 0}(x_n^6 A_n^4 - x_n^2 + 1/3).$$

4.3 Finite summation proof and the boundary cancellation

Write CT_N for the coefficient of $N^0(\log N)^0$ in a finite-cutoff asymptotic expansion. It will be applied only after an exact finite identity has been established. Set

$$T_{d,m} = \mathrm{CT}_N \sum_{k=1}^{N} k^m A_k^d, \qquad b_d(a) = [N^{-a}]A_N^d. \tag{4.19}$$

The coefficients $b_d(a)$ are rational and are fixed by

$$A_N \sim \sum_{a\geq 1} B_{a-1} N^{-a}. \tag{4.20}$$

When a sum converges, its constant term is its ordinary value. In particular, with the descending-index convention for multiple zeta values,

$$T_{1,-p} = \zeta(2,p) \quad (p \geq 1), \qquad T_{0,-p} = \zeta(p) \quad (p \geq 2), \qquad T_{0,-1} = \gamma. \tag{4.21}$$

The following elementary formula will also be used:

$$T_{1,q} = \frac{2-q}{4} B_{q-1} - \frac{1}{2}\delta_{q,1}\gamma, \qquad q \geq 1 \text{ odd}. \tag{4.22}$$

For completeness let $F_q(N) = \sum_{k=1}^{N} k^q$. Finite summation by parts gives

$$\sum_{k=1}^{N} k^q A_k = F_q(N) A_N + \sum_{k=1}^{N} \frac{F_q(k-1)}{k^2}.$$

Insert Faulhaber's polynomial and (4.20). For odd $q \geq 3$, only its quadratic and degree-q terms can contribute nonzero constant coefficients, and they give $(2-q)B_{q-1}/4$. At $q = 1$, the second sum is $(N - H_N)/2$ and the first term has constant coefficient $1/4$, giving $1/4 - \gamma/2$.

The inherited cubic coefficient has the finite-cutoff form

$$C_\ell = 3T_{2,2\ell-2} + 3T_{1,2\ell-4} + T_{0,2\ell-6} + b_3(2\ell). \tag{4.23}$$

This is the cubic summation identity proved in the incoming report. One may verify its explicit evaluation (4.7)–(4.9) by applying the preceding summation by parts once more to the quadratic tail. Its degree-$2\ell-2$ Faulhaber term cancels the explicit linear tail in (4.23); the remaining odd linear powers are evaluated by (4.22). The terms at zero and negative powers give C_1, the $-\zeta(2)/2$ exception at $\ell = 2$, and the boundary contribution $1/4$ at $\ell = 3$. Thus every low-index exception retained in (4.8) is required.

The quartic coordinate corresponding to the odd monomial k^{2j+1} is

$$\widehat{D}_j = 4T_{3,2j-1} + 6T_{2,2j-3} + 4T_{1,2j-5} + T_{0,2j-7} + b_4(2j+1). \tag{4.24}$$

We now prove $\widehat{D}_j = D_j$ with no limit of a divergent series taken before the cancellation.

For $j \geq 1$, put $m = 2j-1$ and

$$F_m(N) = \sum_{k=1}^{N} k^m = \sum_{\ell=1}^{j} f_{2\ell} N^{2\ell} + \frac{1}{2}N^m, \qquad f_{2\ell} = \frac{1}{2j}\binom{2j}{2\ell} B_{2j-2\ell}. \tag{4.25}$$

This includes $j = 1$. Since $F_m(k-1) = F_m(k) - k^m$ and $A_{k-1} = A_k + k^{-2}$, the exact finite identity is

$$\sum_{k=1}^{N} k^m A_k^3 = F_m(N)A_N^3 + \sum_{k=1}^{N} F_m(k-1)\left(\frac{3A_k^2}{k^2} + \frac{3A_k}{k^4} + \frac{1}{k^6}\right). \tag{4.26}$$

Multiply this by four and add the other terms in (4.24). The coefficient $-k^m/2$ in $F_m(k-1)$ cancels the entire quadratic-tail sum $6T_{2,m-2}$. The even Faulhaber powers form precisely four copies of (4.23); their boundary terms cancel the even part of $4F_m(N)A_N^3$. Taking constant coefficients only now yields

$$\widehat{D}_j = 4\sum_{\ell=1}^{j} f_{2\ell} C_\ell + 2b_3(2j-1) + b_4(2j+1) - 2T_{1,2j-5} - T_{0,2j-7}. \tag{4.27}$$

The decisive rational identity is

$$2b_3(2j-1) + b_4(2j+1) = \begin{cases} 0, & j = 1, 2, \\ B_{2j-6}, & j \geq 3. \end{cases} \tag{4.28}$$

Here is a direct proof. In a formal variable $u = N^{-1}$, let $E_0(u) = \sum_{k\geq 0} B_{2k}u^{2k}$. Then $A = uE_0(u) - u^2/2$. The odd parts of the third and fourth powers are

$$(A^3)_{\text{odd}} = u^3E_0(u)^3 + \frac{3}{4}u^5E_0(u), \qquad (A^4)_{\text{odd}} = -2u^5E_0(u)^3 - \frac{1}{2}u^7E_0(u).$$

The E_0^3 terms cancel in the indicated pair of coefficients, leaving $[u^{2j-6}]E_0(u)$. This proves (4.28).

For $j = 1$, (4.21) and the classical evaluation $\zeta(2,3) = 9\zeta(5)/2 - 2\zeta(2)\zeta(3)$ give the correction $-2\zeta(2,3) - \zeta(5) = Q_5$. For $j = 2$, the correction is $-2\zeta(2,1) - \zeta(3) = -3\zeta(3)$. For $j = 3$, (4.22) makes the two Euler constants cancel:

$$1 - 2(1/4 - \gamma/2) - \gamma = \frac{1}{2}.$$

For $j \geq 4$, the last ordinary-power sum has zero constant coefficient, and (4.22) gives

$$B_{2j-6} - \frac{7-2j}{2}B_{2j-6} = \frac{2j-5}{2}B_{2j-6}.$$

Substitution in (4.27) proves (4.10) for every $j \geq 1$.

For $j = 0$, all sums in (4.24) converge and the boundary vanishes. Sorting the four tail indices, or expanding the displayed powers, gives

$$\begin{aligned}\widehat{D}_0 = {} & 24\zeta(2,2,2,1) + 12\zeta(2,2,3) + 12\zeta(2,4,1) + 12\zeta(4,2,1) \\ & + 4\zeta(2,5) + 6\zeta(4,3) + 4\zeta(6,1) + \zeta(7) = Q_7. \end{aligned} \tag{4.29}$$

The last evaluation is Hoffman's explicitly printed quartic example; his sum starts at $n = 1$, so adding $A_0^4 = \zeta(2)^4$ removes his $-175\zeta(8)/24$ term. It is also reconstructed in the supplementary exact shuffle/stuffle verifier.

Finally,

$$\sum_{n=0}^{k-1} x_n^{2r} = \frac{1}{2r+1} \sum_{j=0}^{r} \binom{2r+1}{2j+1} B_{2r-2j}(1/2) k^{2j+1}. \tag{4.30}$$

Every subtracted even-power sum in (4.4) has zero constant coefficient as a polynomial in N. Finite summation by parts using (4.30) therefore gives (4.12). The fixed space follows from (4.8) and (4.10). The first six coefficient rows, provided explicitly in the supplementary data, give (4.14) by rational elimination. This completes the proof.

4.4 An analytic generator with fixed initial constants

For a holomorphic function f define its anchored repeated primitive by

$$(I^k f)(t) = \frac{1}{(k-1)!} \int_0^t (t-u)^{k-1} f(u)\, du, \qquad k \geq 1. \tag{4.31}$$

Paths lie in the disk $|t| < 2\pi$; the integral is path-independent there. The established square and cubic generators are

$$G_2(t) = \frac{t}{2\sinh(t/2)} \left(3\zeta(3) - \frac{1}{4t} \int_0^t (t-u) u^2 \coth(u/2)\, du \right), \tag{4.32}$$

$$G_3(t) = \frac{1}{2\sinh(t/2)} I^2 \left(3\cosh(t/2) G_2(t) - \frac{17}{4}\zeta(4) - \frac{1}{4}\zeta(2) t^2 + \frac{t^4}{96} \right). \tag{4.33}$$

Both formulas are inherited [4, 2]. In particular $2\sinh(t/2)G_3(t) = \sum_{j\geq 1} C_j t^{2j}/(2j)!$.

Theorem 4.3 (Quartic moment generator)**.** *The exponential generating series*

$$G_4(t) = \sum_{r\geq 0} V_r \frac{t^{2r}}{(2r)!} \tag{4.34}$$

converges for $|t| < 2\pi$ and equals

$$\boxed{G_4(t) = \frac{Q_7 t + Q_5 t^3/6 + 4I^2(\cosh(t/2) G_3(t)) - 2I^4(\sinh(t/2) G_2(t))}{2\sinh(t/2)}}. \tag{4.35}$$

The apparent singularity at zero is removable, with $G_4(0) = Q_7$. The quotient is an even holomorphic function on the disk.

Proof. Set

$$D(t) = \sum_{j\geq 0} D_j \frac{t^{2j+1}}{(2j+1)!}, \quad C(t) = 2\sinh(t/2) G_3(t), \quad E(t) = \frac{t}{2}\coth(t/2).$$

The Bernoulli transform (4.12) is exactly $G_4(t) = D(t)/(2\sinh(t/2))$. Summing the convolution in (4.10) gives

$$\begin{aligned} D''(t) &= 4\frac{E(t)C(t)}{t} + Q_5 t - \frac{1}{2}\zeta(3)t^3 + \frac{1}{2}I^4(tE(t)) \\ &= 4\cosh(t/2)G_3(t) + Q_5 t - 2I^2(\sinh(t/2)G_2(t))\,. \end{aligned}$$

To check the last cancellation directly, (4.32) implies

$$\sinh(t/2)G_2(t) = \frac{3}{2}\zeta(3)t - \frac{1}{4}I^2(tE(t)).$$

Thus the cubic polynomial from $-3\zeta(3)$ at $j = 2$ cancels the first term in this last identity after integration. Integrating D'' twice with $D(0) = 0$, $D'(0) = Q_7$ proves (4.35). All the Bernoulli series and inherited generators converge on $|t| < 2\pi$; the anchored integrals preserve holomorphy. The numerator is odd and has first term $Q_7 t$, while the denominator is odd with first term t. This proves the stated parity, value, and removability. No assertion that 2π is the exact Taylor radius is needed here. □

4.5 Finite polylogarithmic form and a rational extraction

Formula (4.35) is an explicit triangular integral generator. It also gives a finite polylogarithm expression, with depth at most three, by the following anchored rule. Put $z = e^{-t}$ and write $L_{\boldsymbol{a}}(t) = \mathrm{Li}_{\boldsymbol{a}}(z)$, with the argument on the outer index:

$$L_{a_1,\dots,a_d}(t) = \sum_{n_1>\cdots>n_d\geq 1} \frac{e^{-tn_1}}{n_1^{a_1}\cdots n_d^{a_d}} \qquad (t>0).$$

The complete inherited cubic expression is reproduced here to make the finite compiler self-contained [2, Section 6, finite depth-two generator]:

$$\begin{aligned} C(t) = {} & \left(9\zeta(4) + 6t\zeta(3) - \frac{t^4}{16}\right)L_2(t) + \left(12\zeta(3) - \frac{t^3}{2}\right)L_3(t) \\ & - 3t^2 L_4(t) - 12tL_5(t) - \frac{45}{2}L_6(t) - \frac{3}{2}t^2 L_{2,2}(t) \\ & - 6t\,(L_{3,2}(t) + L_{2,3}(t)) - 9\,(L_{4,2}(t) + L_{2,4}(t)) - 12L_{3,3}(t) \\ & + \frac{15}{2}\zeta(6) - 6\zeta(3)^2 + 6t\zeta(5) + \frac{1}{2}t^3\zeta(3) + \frac{1}{8}t^2\zeta(4) - \frac{1}{48}t^4\zeta(2) - \frac{t^6}{1440}. \end{aligned} \tag{4.36}$$

Initially this expression is read for real $0 < t < 2\pi$; the integral formula (4.33) specifies its holomorphic continuation through zero and throughout the disk. The derivative identities

$$\frac{d}{dt}L_{a+1,\boldsymbol{b}} = -L_{a,\boldsymbol{b}}, \qquad \frac{d}{dt}L_{1,\boldsymbol{a}} = -\frac{z}{1-z}L_{\boldsymbol{a}}$$

show that multiplication by $z/(1-z)$ raises the needed depth by at most one. The incoming expression for $C(t) = 2\sinh(t/2)G_3(t)$ is a finite sum of polynomials times polylogarithms of depth at most two, plus zeta polynomials, as displayed in (4.36). Since

$$4\cosh(t/2)G_3(t) = 2C(t) + 4\frac{z}{1-z}C(t),$$

the new generator has depth at most three after the two anchored integrations. The remaining term uses only classical polylogarithms:

$$T(t) := 2\sinh(t/2)G_2(t) = 2t\zeta(3) + 3\zeta(4) - \frac{t^4}{48} - \frac{t^2}{2}\mathrm{Li}_2(z) - 2t\,\mathrm{Li}_3(z) - 3\,\mathrm{Li}_4(z). \tag{4.37}$$

Consequently the numerator of (4.35) can alternatively be compiled from the finite expression

$$Q_7 t + \frac{Q_5 t^3}{6} + 2I^2 C + 4I^2 \left(\frac{z}{1-z} C\right) - I^4 T. \tag{4.38}$$

This asserts a sufficient finite function class, not its minimal depth.

Here is an exact endpoint prescription for that compilation. If $F_j' = -F_{j-1}$, an unanchored k-fold primitive of $t^m F_0(t)$ is

$$(-1)^k \sum_{j=0}^{m} \binom{k+j-1}{j} \frac{m!}{(m-j)!} t^{m-j} F_{k+j}(t). \tag{4.39}$$

Subtract its Taylor polynomial of degree $k-1$ at zero to obtain the anchored primitive. For the twofold integration of $zL_{\boldsymbol{a}}/(1-z)$, the only possibly divergent endpoint is the first-index-one value when the polynomial degree is zero. It must be combined before taking a limit. The convergent constant

$$\lambda_{\boldsymbol{a}} = \lim_{t \downarrow 0} \Big(L_{1,\boldsymbol{a}}(t) - \zeta(\boldsymbol{a}) \operatorname{Li}_1(e^{-t}) \Big)$$

is minus the sum of all the convergent terms in the stuffle of (1) with $\boldsymbol{a}$, excluding its leading $(1, \boldsymbol{a})$ term. Indeed, for a nonempty word appearing in C its first index is at least two, so

$$\sum_{n \geq 1} \frac{H_{n-1}^{(\boldsymbol{a})} - \zeta(\boldsymbol{a})}{n}$$

converges absolutely, and dominated convergence identifies it with the displayed limit. The finite harmonic stuffle evaluates this sum, since $H_N(H_N^{(\boldsymbol{a})} - \zeta(\boldsymbol{a})) \to 0$. Here $H_N^{(\boldsymbol{a})}$ uses the same strict descending-index convention as the multiple zeta values. Pure zeta-polynomial terms in (4.36) are included by adjoining the empty word: $L_\varnothing = 1$, $\zeta(\varnothing) = 1$, $L_{1,\varnothing} = \operatorname{Li}_1(e^{-t})$, and $\lambda_\varnothing = 0$. In particular, the degree-zero empty-word coefficient is $15\zeta(6)/2 - 6\zeta(3)^2$. With this convention the degree-zero coefficients in C satisfy $\sum c_{0,\boldsymbol{a}} \zeta(\boldsymbol{a}) = C(0) = 0$. Thus all the divergent Li_1 endpoints cancel in (4.38), leaving exactly the prescribed $\lambda_{\boldsymbol{a}}$ combination. This rule fixes the integration constants rather than leaving an undetermined polynomial.

The six-coordinate theorem also produces ordinary rational evaluations. For example set

$$\begin{aligned} P(x) &= 5233 + 697660x^2 + 9382304x^4 + 28113792x^6 \\ &\quad + 21816576x^8 + 3886080x^{10}, \\ R(x) &= 3886080x^6 + 20521216x^4 + 21456896x^2 + \frac{940103168}{315}. \end{aligned}$$

Then

$$\boxed{\sum_{n=0}^{\infty} \big(P(x_n)\psi'(n+1)^4 - R(x_n)\big) = -\frac{775936}{3}.} \tag{4.40}$$

Its summand is $O(x_n^{-2})$. This is an exact rational elimination of the six proved coefficient rows, with the same polynomial subtractions as (4.4).

4.6 Verification and the remaining question

The supplementary program reconstructs each finite-cutoff coordinate without using (4.10). It recursively applies Faulhaber's polynomial and $A_{k-1} = A_k + k^{-2}$, and expands the convergent negative-power tail sums into ordinary multiple zeta values. A separate shuffle/stuffle and duality computation reduces weights through seven. It agrees with (4.10) for $0 \leq j \leq 24$, checks the boundary cancellation at 24 indices, and reconstructs the first 20 centered moments independently.

It also verifies 24 generator coefficients, the stated six-coordinate determinant, and the rational identity (4.40) by exact arithmetic.

For a separate numerical diagnostic, the defining finite sums in (4.4) were evaluated at 120 decimal working digits and supplemented by centered Hurwitz tails. The two choices $(N, J) = (120, 36)$ and $(240, 54)$ use N direct summands and retain J asymptotic coefficients. For $0 \le r \le 8$, the largest second-cutoff residual against the closed formulas is below 3.6×10^{-90}. The data record both cutoffs and the residuals. These are numerical diagnostics, not interval certificates; the finite summation proof establishes the all-index result.

The quartic case of the stated incoming question is therefore answered. For fifth and higher powers, the useful next target is an analogue of (4.28) that identifies all surviving lower-tail terms before constants are taken. It should determine a fixed coordinate space and a generator at each power. The present quartic theorem does not assert that the same six constants suffice for those powers.

5 Several quadratic factors beyond the first spectral derivative

The incoming quadratic-spectrum report [5] proves additivity of the first spectral derivative for any even polynomial weight and any finite list of masses. Its explicit next question asks for the second derivative, the logarithmic cross term, and the limit when two masses coincide. The following results answer that question with an ordinary convergent series and a one-dimensional integral of elementary functions. They also identify the finite corrections needed at every higher spectral order. The classical multiplicative-anomaly framework [11] concerns first derivatives of spectral zeta functions; the statements below concern the additional higher derivatives. No historical-priority claim is needed for this continuation of the inspected repository.

Fix $0 < a < 1$, put $v_n = (n + a)^2$ for $n \in \mathbb{Z}$, and let

$$P(v) = \sum_{d=0}^{D} p_d v^d, \qquad B_j > 0 \quad (1 \le j \le r), \qquad r \ge 2.$$

The parameter B_j is the squared mass. Initially in their respective half-planes of absolute convergence, define

$$\mathcal{E}_P(s; a, B) = \sum_{n\in\mathbb{Z}} P(v_n)(v_n + B)^{-s}, \tag{5.1}$$

$$\mathcal{Z}_{P,\boldsymbol{B}}(s; a) = \sum_{n\in\mathbb{Z}} P(v_n) \prod_{j=1}^{r} (v_n + B_j)^{-s}. \tag{5.2}$$

Their continuations are holomorphic at $s = 0$. Indeed, a tail expansion uses integer powers of v_n^{-1}, and a possible pole at zero would require an odd power $(n + a)^{-1}$. The detailed subtraction proof below also supplies this continuation directly.

Write

$$\mathfrak{D}_k(P; a, \boldsymbol{B}) = r^{k-1} \sum_{j=1}^{r} \mathcal{E}_P^{[k]}(0; a, B_j) - \mathcal{Z}_{P,\boldsymbol{B}}^{[k]}(0; a), \qquad k \ge 1, \tag{5.3}$$

where square brackets indicate derivatives in the spectral parameter. The factors r^{k-1} are part of the definition. They result from comparing the factors at the same spectral scale:

$$\mathfrak{D}(s) = \frac{1}{r} \sum_{j} \mathcal{E}_P(rs; a, B_j) - \mathcal{Z}_{P,\boldsymbol{B}}(s; a), \qquad \mathfrak{D}_k = \mathfrak{D}^{[k]}(0). \tag{5.4}$$

The already established first-derivative identity is $\mathfrak{D}_1 = 0$.

5.1 An exact second-derivative formula

For a function with a Laurent expansion at $v=\infty$, let Pol_∞ retain its nonnegative integer powers of v. Define

$$T_P(v;B,C)=\mathrm{Pol}_\infty\left\{P(v)\log^2\frac{v+B}{v+C}\right\}, \tag{5.5}$$

$$S_P(a;B,C)=\sum_{n\in\mathbb{Z}}\left\{P(v_n)\log^2\frac{v_n+B}{v_n+C}-T_P(v_n;B,C)\right\}. \tag{5.6}$$

Every term in the subtraction is explicit. In fact,

$$q_k(B,C)=(-1)^k\sum_{m=1}^{k-1}\frac{(B^m-C^m)(B^{k-m}-C^{k-m})}{m(k-m)},\qquad k\ge 2, \tag{5.7}$$

$$T_P(v;B,C)=\sum_{d=2}^{D}p_d\sum_{k=2}^{d}q_k(B,C)v^{d-k}. \tag{5.8}$$

Empty sums are zero. Thus the weights $P(v)=1$ and $P(v)=v$ need no subtraction. For $P(v)=v^2$ the only subtraction is $(B-C)^2$, while for $P(v)=v^3$ it is

$$(B-C)^2v-(B-C)^2(B+C).$$

Theorem 5.1 (Second spectral derivative for any number of masses)**.** *The series* (5.6) *is ordinary and absolutely convergent, and*

$$\boxed{\mathfrak{D}_2(P;a,\boldsymbol{B})=\sum_{1\le i<j\le r}S_P(a;B_i,B_j).} \tag{5.9}$$

Consequently

$$\mathcal{Z}^{[2]}_{P,\boldsymbol{B}}(0;a)=r\sum_j\mathcal{E}^{[2]}_P(0;a,B_j)-\sum_{i<j}S_P(a;B_i,B_j). \tag{5.10}$$

For $\deg P\le 1$*, the pair correction is the unsubtracted convergent sum of logarithm squares. In particular, for* $P=1$ *it is nonzero whenever at least two masses are distinct. Thus first-derivative additivity does not extend unchanged to second derivatives.*

Proof. At a fixed $v>0$ write $L_j=\log(v+B_j)$. Twice differentiating the common-scale summand at zero gives

$$P(v)\left\{r\sum_j L_j^2-\left(\sum_j L_j\right)^2\right\}=P(v)\sum_{i<j}(L_i-L_j)^2.$$

The logarithm difference has the expansion

$$\log\frac{v+B}{v+C}=\sum_{m\ge1}\frac{(-1)^{m+1}(B^m-C^m)}{mv^m}$$

on a sufficiently large tail. Squaring proves (5.7) and (5.8); the subtracted summand is $O(v^{-1})$, uniformly for masses in a compact positive set. It therefore has an absolutely convergent bilateral sum.

It remains to retain the exact value of the removed polynomial. Put

$$\mathcal{H}_a(s)=\zeta(2s,a)+\zeta(2s,1-a). \tag{5.11}$$

For every integer $m\ge 0$,

$$\mathcal{H}_a(-m)=0, \tag{5.12}$$

because $\zeta(-2m,x) = -B_{2m+1}(x)/(2m+1)$ and $B_{2m+1}(1-a) = -B_{2m+1}(a)$. Hence each removed v^m has zero bilateral spectral value. The common-scale normal-convergence argument in Theorem 5.4 makes this differentiation and restoration precise and gives zero finite correction at order two. This proves (5.9). With an affine weight, the raw tail is $O(v^{-1})$ or better. With $P = 1$ every pair summand is nonnegative and a distinct pair has strictly positive summands, giving the final claim. □

For two masses, the mixed logarithmic coordinate is equivalently

$$\operatorname{Reg}\sum_{n\in\mathbb{Z}} P(v_n)\log(v_n+B)\log(v_n+C) = \frac{\mathcal{E}_P^{[2]}(0;a,B)+\mathcal{E}_P^{[2]}(0;a,C)-S_P(a;B,C)}{2}. \tag{5.13}$$

Here Reg denotes the analytic continuation determined by the common quadratic spectral coordinate. Formula (5.6), or the ordinary integral below, makes the new cross-term correction explicit; it does not subtract two unspecified divergent sums.

5.2 An elementary integral for the entire correction

The classical bilateral resolvent, from cotangent partial fractions [13], is

$$R_a(t) = \sum_{n\in\mathbb{Z}} \frac{1}{v_n+t} = \frac{\pi}{\sqrt{t}}\frac{\sinh(2\pi\sqrt{t})}{\cosh(2\pi\sqrt{t})-\cos(2\pi a)}, \qquad t>0. \tag{5.14}$$

It follows by pairing the cotangent partial fractions at $a+i\sqrt{t}$ and $a-i\sqrt{t}$. Set

$$f_P(t) = P(-t)R_a(t).$$

Theorem 5.2 (Ordinary logarithmic quadrature)**.** *For $B > C > 0$,*

$$\boxed{S_P(a;B,C) = 2\int_C^B f_P(t)\log\frac{B-t}{t-C}\,\mathrm{d}t.} \tag{5.15}$$

The integral is ordinary and absolutely convergent; its only endpoint singularities are logarithmic. For arbitrary positive B, C, use $S_P(a;B,C) = S_P(a;C,B)$ and $S_P(a;B,B) = 0$. A form with a continuous integrand on the square is

$$S_P(a;B,C) = \int_C^B\int_C^B \frac{f_P(t)-f_P(u)}{u-t}\,\mathrm{d}u\,\mathrm{d}t, \tag{5.16}$$

where the diagonal value is $-f_P'(t)$.

Proof. Polynomial division gives

$$P(v)/(v+t) = Q_t(v) + P(-t)/(v+t)$$

with a polynomial Q_t. Therefore subtraction of the polynomial part and summation of the rational remainder give

$$\sum_{n\in\mathbb{Z}}(1-\mathrm{Pol}_\infty)\frac{P(v_n)}{(v_n+t)(v_n+u)} = \frac{P(-t)R_a(t)-P(-u)R_a(u)}{u-t}.$$

The statement at $u = t$ follows by continuity. For (t,u) in a compact positive square, every subtracted summand is uniformly $O(v_n^{-1})$. Normal convergence permits integration of this identity over $[C,B]^2$. Since

$$\int_C^B \frac{\mathrm{d}t}{v+t} = \log\frac{v+B}{v+C},$$

and finite polynomial extraction commutes with these integrals, the left side becomes (5.6). This proves (5.16).

To reduce the square integral, omit temporarily the strip $|u-t| \le \varepsilon$, split the numerator, and interchange u, t in its second term. The result tends to

$$2\int_C^B f_P(t)\big(\log(B-t) - \log(t-C)\big)\,\mathrm{d}t.$$

The omitted-strip contribution tends to zero because the divided difference extends continuously to the diagonal. The endpoint logarithms are integrable. Thus the temporary symmetric exclusion is only a proof device; the final integral needs no principal value or finite-part convention. □

For example, at $a = 1/2$, $P = 1$, $B = 4$, $C = 1$, the exact correction is

$$\sum_{n\in\mathbb{Z}} \log^2 \frac{(n+1/2)^2+4}{(n+1/2)^2+1} = 2\pi \int_1^4 \frac{\tanh(\pi\sqrt{t})}{\sqrt{t}} \log\frac{4-t}{t-1}\,\mathrm{d}t. \tag{5.17}$$

At the quartic weight $P(v) = v^2$, the corresponding exact identity is

$$\sum_{n\in\mathbb{Z}} \left[(n+1/2)^4 \log^2 \frac{(n+1/2)^2+4}{(n+1/2)^2+1} - 9\right] = 2\pi \int_1^4 t^{3/2}\tanh(\pi\sqrt{t}) \log\frac{4-t}{t-1}\,\mathrm{d}t. \tag{5.18}$$

Both series and both integrals are ordinary convergent expressions.

5.3 Every confluent coefficient in finite resolvents

Let $H_m = \sum_{j=1}^m 1/j$ and $H_0 = 0$, and put $\rho = \min(a, 1-a)$. For $k \ge 1$, define the rationally subtracted resolvent moment

$$R_{P,k}(a;C) = \sum_{n\in\mathbb{Z}} (1 - \mathrm{Pol}_\infty) \frac{P(v_n)}{(v_n + C)^k}. \tag{5.19}$$

As above, this is an absolutely convergent series after its explicit polynomial division.

Theorem 5.3 (Complete confluent expansion)**.** *For $C > 0$ and $|h| < C + \rho^2$,*

$$S_P(a; C+h, C) = 2\sum_{k=2}^{\infty} \frac{(-1)^k H_{k-1}}{k} h^k R_{P,k}(a;C), \tag{5.20}$$

$$= -2\sum_{k=2}^{\infty} \frac{H_{k-1}}{k!} f_P^{(k-1)}(C) h^k. \tag{5.21}$$

Here $S_P(a; C+h, C)$ denotes its principal-logarithm analytic continuation from positive masses to this disk. Indeed every $v_n + C + h$ has positive real part on the disk, so the logarithms and the explicitly subtracted series define a single holomorphic function there. Each coefficient has the finite reduction

$$R_{P,k}(a;C) = \frac{(-1)^{k-1}}{(k-1)!} f_P^{(k-1)}(C) = \sum_{\ell=0}^{\min(D,k-1)} \frac{P^{(\ell)}(-C)}{\ell!} \mathcal{E}_1(k-\ell; a, C). \tag{5.22}$$

The values $\mathcal{E}_1(N; a, C)$ at positive integers are obtained by differentiating the elementary function (5.14), or by the manuscript's finite polygamma partial fractions.

Proof. The elementary Taylor identity

$$\log^2(1+x) = 2\sum_{k=2}^{\infty} \frac{(-1)^k H_{k-1}}{k} x^k, \qquad |x| < 1,$$

follows by differentiating both sides and using $\log(1+x)/(1+x) = \sum_{m\geq 1}(-1)^{m+1} H_m x^m$. Apply it with $x = h/(v_n + C)$, extract the polynomial parts before summation, and use locally uniform convergence on the stated disk. This proves (5.20).

Differentiating $f_P(C)$ under the subtracted resolvent sum proves the first equality in (5.22). For the second, expand $P(v)$ at $v = -C$. Terms with $\ell \geq k$ are polynomials and disappear under $1 - \mathrm{Pol}_\infty$; the other terms give the displayed finite sum. The derivative formula yields (5.21). The function f_P is meromorphic in C, with possible poles only at $C = -v_n$. The nearest possible pole is at distance $C + \rho^2$, which also proves the claimed disk of convergence. Zeros of P can only remove poles and enlarge this disk. □

The first terms are

$$S_P(a; C+h, C) = h^2 R_{P,2}(a; C) - h^3 R_{P,3}(a; C) + \frac{11}{12} h^4 R_{P,4}(a; C) + O(h^5).$$

In particular, for $P = 1$, $a = 1/2$, and $C = 1$,

$$S_1(1/2; 1+h, 1) = \frac{1}{2}\left(\pi \tanh \pi - \pi^2 \operatorname{sech}^2 \pi\right) h^2 + O(h^3). \tag{5.23}$$

This explicitly evaluates the first nonzero coefficient when two quadratic factors merge.

5.4 A finite correction formula at every spectral order

The vanishing (5.12) removes polynomial counterterms at the second derivative. Higher derivatives also produce polynomials times powers of $\log v$, whose spectral values need not vanish. The following formula retains every such contribution.

Define

$$A(s,z) = \frac{1}{r}\sum_{j=1}^{r}(1+B_j z)^{-rs} - \prod_{j=1}^{r}(1+B_j z)^{-s} = \sum_{q\geq 2} a_q(s) z^q, \tag{5.24}$$

$$a_q(s) = (-1)^q \left\{ \frac{1}{r}\sum_{j=1}^{r} \frac{(rs)_q B_j^q}{q!} - \sum_{q_1+\cdots+q_r=q} \prod_{j=1}^{r} \frac{(s)_{q_j} B_j^{q_j}}{q_j!} \right\}. \tag{5.25}$$

The coefficient polynomials $a_q(s)$ are divisible by s^2; the coefficients a_0, a_1 vanish identically. Put

$$T(s,v) = \sum_{d=2}^{D} p_d \sum_{q=2}^{d} a_q(s) v^{d-q}. \tag{5.26}$$

Theorem 5.4 (All-order ordinary remainder and finite Hurwitz jets)**.** *For s in a neighborhood of zero,*

$$\begin{aligned}\mathfrak{D}(s) = &\sum_{n\in\mathbb{Z}} v_n^{-rs}\left(P(v_n)A(s, v_n^{-1}) - T(s, v_n)\right) \\ &+ \sum_{d=2}^{D} p_d \sum_{q=2}^{d} a_q(s)\mathcal{H}_a(rs+q-d).\end{aligned} \tag{5.27}$$

The first series is normally convergent near zero. For $k \geq 2$, define

$$F_k(v) = (-1)^k P(v) \left\{ r^{k-1} \sum_j \log^k(v + B_j) - \left(\sum_j \log(v + B_j) \right)^k \right\}, \tag{5.28}$$

$$T_k(v) = \sum_{d=2}^{D} p_d \sum_{q=2}^{d} \sum_{j=2}^{k} \binom{k}{j} a_q^{[j]}(0) (-r \log v)^{k-j} v^{d-q}, \tag{5.29}$$

$$C_k = \sum_{d=2}^{D} p_d \sum_{q=2}^{d} \sum_{j=2}^{k-1} \binom{k}{j} a_q^{[j]}(0) r^{k-j} \mathcal{H}_a^{[k-j]}(q-d). \tag{5.30}$$

Then the exact kth spectral defect is

$$\boxed{\mathfrak{D}_k = \sum_{n \in \mathbb{Z}} (F_k(v_n) - T_k(v_n)) + C_k.} \tag{5.31}$$

The remainder series is ordinary and absolutely convergent. Its summand is $O(v_n^{-1}(1 + |\log v_n|^{k-2}))$. All correction terms are finite Hurwitz-zeta jets at nonpositive even integers. For affine P, both T_k and C_k vanish at every $k \geq 2$.

Proof. At $s = 0$, the two terms in (5.24) both equal one, and their first s derivatives both equal $-\sum_j \log(1 + B_j z)$. Thus $a_q(s) = O(s^2)$. Expansion to first degree in z shows that $a_1 = 0$. The Pochhammer expansion proves (5.25) exactly.

On a sufficiently large lattice tail, Taylor expansion in $z = v^{-1}$ and the definition of T give

$$P(v)A(s, v^{-1}) - T(s, v) = O(v^{-1})$$

uniformly for s in a compact neighborhood of zero. Multiplication by v^{-rs} leaves a normally convergent series, for example on $|\operatorname{Re} s| < 1/(4r)$. Finitely many smaller v_n cause no difficulty: the exact functions $A(s, v_n^{-1})$ are used there, without applying a possibly invalid small-z expansion.

Restoring the removed monomials by $\sum_n v_n^{d-q-rs} = \mathcal{H}_a(rs + q - d)$ gives (5.27), first in a common half-plane after a finite head is separated and then by meromorphic continuation. Alternatively, the same head-tail construction is a direct definition of this continuation. None of the restored factors has a pole at zero: $q - d$ is an integer, whereas the only pole of $\mathcal{H}_a$ is at $1/2$. This proves the asserted holomorphy and normal convergence.

Differentiate (5.27) k times. The first raw derivative is (5.28), and the derivative of $v^{-rs}T(s, v)$ is (5.29). In the restored finite sum, the terms with $j = 0, 1$ vanish because a_q is divisible by s^2; the term $j = k$ vanishes by (5.12). The remaining terms are exactly (5.30). Derivatives of the normal tail give the stated polynomial-logarithmic bound. For $D \leq 1$, all finite sums are empty. □

The second-derivative theorem follows immediately: at $k = 2$ the finite sum C_2 is empty, the raw derivative is the pairwise square, and T_2 is precisely its polynomial part. At the third derivative the correction can already be nonzero. For two masses and $P(v) = v^2$,

$$F_3(v) = -3v^2 \log((v + B)(v + C)) \log^2 \frac{v + B}{v + C}, \tag{5.32}$$

$$T_3(v) = -6(B - C)^2 \log v, \tag{5.33}$$

$$C_3 = -12(B - C)^2 \log(2 \sin \pi a). \tag{5.34}$$

Indeed $a_2(s) = \frac{1}{2}(B - C)^2 s^2$ in this case, while Lerch's identity and Gamma reflection give

$$\mathcal{H}_a'(0) = 2(\zeta'(0, a) + \zeta'(0, 1 - a)) = -2 \log(2 \sin \pi a).$$

Formula (5.34) is therefore a concrete check on the sign and the scale factor in the general correction law. Discarding it would change the third spectral derivative.

5.5 Polylogarithmic coordinates for the finite corrections

Define the symmetric polylogarithm

$$\mathcal{C}_s(a) = \frac{\mathrm{Li}_s(e^{2\pi i a}) + \mathrm{Li}_s(e^{-2\pi i a})}{2}.$$

For real s this is the real part of the first polylogarithm; the symmetric definition retains holomorphic dependence on complex s. The symmetric Hurwitz functional equation gives, for $m \geq 0$ and u near zero,

$$\mathcal{H}_a(-m+u) = 4(-1)^m \frac{\Gamma(2m+1-2u)}{(2\pi)^{2m+1-2u}} \sin(\pi u) \mathcal{C}_{2m+1-2u}(a). \tag{5.35}$$

Consequently $\mathcal{H}_a^{[h]}(-m)$ is a finite combination of polylogarithm order derivatives at $2m+1$, of order at most $h-1$, and Gamma derivatives at $2m+1$. The zero of $\sin(\pi u)$ is responsible for this reduction in order. For example,

$$\mathcal{H}_a'(-m) = \frac{2(-1)^m (2m)!}{(2\pi)^{2m}} \mathcal{C}_{2m+1}(a). \tag{5.36}$$

Thus the finite corrections in (5.30) use polylogarithm order derivatives only through order $k-3$. This is a finite reduction of the correction, not a claim that the ordinary remainder in (5.31) always reduces to finitely many polylogarithms.

5.6 A confluent law simultaneously at every spectral order

Theorem 5.5 (All-order mass-coalescence law)**.** *Let $B > 0$ and fixed real numbers $c_1, \ldots, c_r$ be given, and set $B_j = B + \varepsilon c_j$. For every $k \geq 2$,*

$$\mathfrak{D}_k(P; a, \boldsymbol{B}) = \binom{k}{2} r^{k-2} \varepsilon^2 \left(r \sum_j c_j^2 - \Big(\sum_j c_j \Big)^2 \right) \mathcal{E}_P^{[k-2]}(2; a, B) + O(\varepsilon^3). \tag{5.37}$$

The right side is well defined even when its defining raw series at $s = 2$ diverges: $\mathcal{E}_P(s; a, B)$ has no pole at an integer. For $\deg P \leq 1$ it is an ordinary spectral derivative of a convergent resolvent series. If all c_j are equal, the masses remain identical and every defect vanishes exactly.

Proof. For a fixed v, expand the common-scale expression in (5.4) through second order in ε. The constant and linear coefficients cancel. The quadratic term is

$$\frac{s^2 \varepsilon^2}{2} \left(r \sum_j c_j^2 - \Big(\sum_j c_j \Big)^2 \right) (v+B)^{-rs-2}.$$

The same tail subtraction as above justifies this identity for the continued families, locally uniformly in s near zero. Hence

$$\mathfrak{D}(s) = \frac{s^2 \varepsilon^2}{2} \left(r \sum_j c_j^2 - \Big(\sum_j c_j \Big)^2 \right) \mathcal{E}_P(rs+2; a, B) + O(\varepsilon^3).$$

Taking its kth derivative gives (5.37). The single-mass tail expansion has possible poles only at half-integers, which proves the assertion at $s = 2$. When all masses are equal, $\mathcal{Z}_{P,\boldsymbol{B}}(s; a) = \mathcal{E}_P(rs; a, B)$, so (5.4) is identically zero. □

5.7 A global Hurwitz-jet evaluation of the individual terms

For completeness, the individual second derivatives in (5.10) can be evaluated without leaving an unevaluated multivariate spectral zeta function. Choose an integer $N \geq 0$ such that $B < (N+\rho)^2$, and define

$$\mathcal{H}_{a,N}(s) = \zeta(2s, N+a) + \zeta(2s, N+1-a).$$

Separating the $2N$ lattice terms with positive coordinates $n+a$ and $n+1-a$, $0 \leq n < N$, gives the geometrically convergent formula

$$\begin{aligned}\mathcal{E}_P^{[2]}(0;a,B) = &\sum_{n=0}^{N-1} \sum_{x\in\{n+a,n+1-a\}} P(x^2)\log^2(x^2+B) \\ &+ \sum_{d=0}^{D} p_d \Bigg\{ \mathcal{H}_{a,N}^{[2]}(-d) \\ &\qquad + 2\sum_{q=1}^{\infty} \frac{(-1)^q B^q}{q} \Big[H_{q-1}\mathcal{H}_{a,N}(q-d) + \mathcal{H}'_{a,N}(q-d) \Big] \Bigg\}. \end{aligned} \tag{5.38}$$

Every spectral argument here is an integer in the $\mathcal{H}$ variable, so no pole is substituted separately. This identity follows directly from

$$(s)_q = (q-1)!s\big(1 + H_{q-1}s + O(s^2)\big)$$

and the binomial expansion of $(x^2+B)^{-s}$ on the chosen tail. The strict tail-radius condition justifies spectral differentiation term by term. Higher derivatives use the same finite Pochhammer coefficient calculus, or unsigned Stirling numbers of the first kind. Together with (5.15), this supplies a global exact evaluation of the several-mass second derivative in ordinary Hurwitz jets and an explicitly anchored elementary integral.

5.8 Verification and the remaining boundary

The accompanying Python program constructs the Pochhammer spectral coefficients and the direct logarithmic coefficients independently over the rational numbers. It compares the resulting ordinary remainders with independent finite-head Hurwitz continuations, evaluates the resolvent quadrature directly, checks the nonzero third-order correction, and compares the convergent confluent series with its integral. Numerical diagnostics are distinct from the all-order proofs above; they are not interval certificates.

The answered repository question is the several-mass second derivative and its confluent limit, with the ordinary logarithmic cross term made explicit. The one-dimensional integral in (5.15) is itself an exact evaluation coordinate. A finite reduction of that integral to a specified algebra of ordinary polylogarithms for arbitrary a, B, C remains a separate question. For odd or noninteger spectral weights, resonant odd powers can occur and additional local Laurent terms must be retained; the parity-protected bilateral formula above does not silently assert their absence.

6 Every spectral derivative of a product of quadratic factors

The preceding quadratic-spectrum report asks both for higher factorization derivatives and for the logarithmic cross term for several masses [5, Section 7, questions 2 and 7]. We give a finite subtraction formula at every derivative order, and then state its conjugate-linear specialization to the precise defect in question 2. Its one-sided version also identifies exactly which generalized Stieltjes constants occur. First-derivative multiplicative anomalies are classical [11, 12]; the result here concerns the specified higher derivatives and their explicit normalization.

Fix $a > 0$, an integer $r \geq 2$, nonnegative real numbers $B_1, \dots, B_r$, and a polynomial $W(x) = \sum_{d=0}^{D} w_d x^d$. Write $x_n = n + a$ and $Q_j(x) = x^2 + B_j$. In their initial half-planes of absolute convergence set

$$E_{W,j}(s;a) = \sum_{n\geq 0} W(x_n) Q_j(x_n)^{-s}, \tag{6.1}$$

$$Z_{W,\boldsymbol{B}}(s;a) = \sum_{n\geq 0} W(x_n) \prod_{j=1}^{r} Q_j(x_n)^{-s}. \tag{6.2}$$

For the quadratic factors Q_j, logarithms are real logarithms of positive numbers. The conjugate-linear specialization below explicitly uses principal complex logarithms. The normalized difference is

$$\mathfrak{D}_W(s) = \frac{1}{r}\sum_{j=1}^{r} E_{W,j}(rs;a) - Z_{W,\boldsymbol{B}}(s;a), \qquad \Delta_m(W) = \mathfrak{D}_W^{(m)}(0). \tag{6.3}$$

Thus $\Delta_m = r^{m-1}\sum_j E_{W,j}^{(m)}(0) - Z_{W,\boldsymbol{B}}^{(m)}(0)$. The factor r^{m-1} is part of the definition, and cannot be omitted when comparing different total spectral degrees.

6.1 A finite coefficient algebra

Define polynomials $h_k(s)$ by the Taylor expansion at $t = 0$,

$$h(s,t) = \frac{1}{r}\sum_{j=1}^{r}(1+B_j t)^{-rs} - \prod_{j=1}^{r}(1+B_j t)^{-s} = \sum_{k\geq 2} h_k(s) t^k. \tag{6.4}$$

An entirely finite formula for each coefficient is

$$h_k(s) = (-1)^k \left\{ \frac{(rs)_k}{r\,k!}\sum_j B_j^k - \sum_{k_1+\cdots+k_r=k}\prod_{j=1}^{r}\frac{(s)_{k_j}B_j^{k_j}}{k_j!} \right\}. \tag{6.5}$$

Here $(s)_0 = 1$. Put $A_{k,j} = h_k^{(j)}(0)$; it is zero for $j > k$. Both h_0 and h_1 vanish identically, and every h_k is divisible by s^2. The latter assertion follows before coefficient extraction: $h(0,t) = \partial_s h(0,t) = 0$. In particular, if

$$V(\boldsymbol{B}) = \sum_{i<j}(B_i - B_j)^2, \tag{6.6}$$

then $h_2(s) = V(\boldsymbol{B})s^2/2$.

Theorem 6.1 (Finite subtraction at all spectral orders)**.** *Choose any integer*

$$K \geq \max\left(1, \left\lfloor \frac{D+1}{2} \right\rfloor\right). \tag{6.7}$$

The functions in (6.3) *are holomorphic at zero, and*

$$\mathfrak{D}_W(s) = \mathcal{R}_K(s) + \sum_{d=0}^{D} w_d \sum_{k=2}^{K} h_k(s)\zeta(2rs+2k-d,a), \tag{6.8}$$

$$\mathcal{R}_K(s) = \sum_{n\geq 0} W(x_n) x_n^{-2rs}\left\{ h(s,x_n^{-2}) - \sum_{k=2}^{K} h_k(s) x_n^{-2k} \right\}. \tag{6.9}$$

The remainder is holomorphic in $\Re s > (D-2K-1)/(2r)$*, with normally convergent fixed spectral derivatives there. Every apparent pole of the finite sum at* $s = 0$ *is removable after multiplication by* $h_k(s)$*.*

For $m \geq 1$ define

$$L_j(x) = \log(x^2 + B_j), \tag{6.10}$$

$$T_m(x) = (-1)^m \left\{ r^{m-1} \sum_j L_j(x)^m - \left(\sum_j L_j(x) \right)^m \right\}, \tag{6.11}$$

$$U_{k,m}(X) = \sum_{j=2}^{m} \binom{m}{j} A_{k,j} (-2rX)^{m-j}. \tag{6.12}$$

An empty sum is zero. Then the following is an ordinary convergent series plus a finite, explicit correction:

$$\boxed{\begin{aligned} \Delta_m(W) = & \sum_{n \geq 0} W(x_n) \left\{ T_m(x_n) - \sum_{k=2}^{K} x_n^{-2k} U_{k,m}(\log x_n) \right\} \\ & + \sum_{d=0}^{D} w_d \sum_{k=2}^{K} \mathcal{C}_{m,k}(2k-d; a), \end{aligned}} \tag{6.13}$$

where, if $\nu \neq 1$,

$$\mathcal{C}_{m,k}(\nu; a) = \sum_{j=2}^{m} \binom{m}{j} A_{k,j} (2r)^{m-j} \zeta^{(m-j)}(\nu, a), \tag{6.14}$$

and at the resonant value $\nu = 1$,

$$\boxed{\mathcal{C}_{m,k}(1; a) = \frac{A_{k,m+1}}{2r(m+1)} + \sum_{j=2}^{m} \binom{m}{j} A_{k,j} (-2r)^{m-j} \gamma_{m-j}(a).} \tag{6.15}$$

Consequently only $\gamma_0(a), \ldots, \gamma_{m-2}(a)$ can occur at the mth derivative. They occur only through odd weight monomials x^{2k-1} with $k \geq 2$.

Proof. For sufficiently large $\Re s$, subtract the finite Taylor polynomial in (6.4) from each summand in (6.3); the removed powers sum to the Hurwitz functions in (6.8). This proves the equality where all series converge absolutely.

On a compact set of s and masses, the Taylor remainder is $O(x^{-2K-2})$ as $x \to \infty$, uniformly with every fixed spectral derivative. A derivative of x^{-2rs} introduces only a power of $\log x$. Hence the summands of (6.9) are bounded by $Cx^{D-2K-2-2r\Re s}(1 + \log x)^M$, which is summable in the stated half-plane. This argument uses the Taylor expansion only for large x. The finitely many small x_n are kept in the original expression, so no assumption such as $B_j < a^2$ is needed. It proves normal convergence, holomorphy, and the legitimacy of termwise differentiation. It also continues the original identity from its initial half-plane to a neighborhood of zero.

The same finite-subtraction argument applied separately to $\prod_j (1 + B_j t)^{-s}$ and $(1 + B_j t)^{-s}$ proves that the original spectral functions are holomorphic at zero. Their positive-t-degree coefficients vanish at $s = 0$ and thus cancel any simple Hurwitz pole; their degree-zero terms are $\zeta(-d, a)$, regular at zero.

The only finite term in (6.8) with a pole at zero has $2k - d = 1$. The Stieltjes convention is

$$\zeta(1 + u, a) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(a) u^j. \tag{6.16}$$

Since $h_k(s) = O(s^2)$, the product is holomorphic. At a nonresonant integer, the ordinary product rule gives (6.14). At a resonance, the pole is $1/(2rs)$, so its mth coefficient contributes $A_{k,m+1}/(2r(m+1))$. The regular coefficients in (6.16) give the remaining terms of (6.15). This proves the finite correction exactly, including its sign and the extra local term.

Finally, differentiating the actual summands in the remainder gives (6.11) and (6.12). The normal convergence just proved allows the differentiation, and yields (6.13). Increasing K merely transfers an exactly summed term between the remainder and the correction, so the answer is independent of that choice. □

6.2 The local first derivative and the nonlocal second derivative

The first derivative is particularly short:

Corollary 6.2 (First-derivative residue)**.** *For the convention* (6.3)*,*

$$\Delta_1(W) = \frac{1}{4r}[x^{-1}]\,W(x)\sum_{i<j}\left\{\log\frac{x^2+B_i}{x^2+B_j}\right\}^2. \tag{6.17}$$

It is independent of a. If $\deg W \le 2$, it vanishes for all masses. For $W = x^3$, it equals $V(\boldsymbol{B})/(4r)$.

Proof. The remainder and nonresonant corrections start at order s^2. Moreover $\partial_s^2 h(0,t)$ equals the sum of the pairwise squared logarithmic differences in (6.17). Substitute $m = 1$ in (6.15). □

At the second derivative the pairwise logarithmic squares are still present, but are no longer purely local. In addition, at resonances one must retain $A_{k,3}/(6r)$. The next examples make both features explicit.

Corollary 6.3 (A single Stieltjes coefficient at every derivative)**.** *Take $r = 2$, $W = x^3$, $B \neq C$, and set $\delta = B - C$. For every integer $m \ge 2$, put*

$$c_m = \binom{m}{2}(-4)^{m-2}\delta^2. \tag{6.18}$$

With T_m as in (6.11)*,*

$$\boxed{\Delta_m(x^3) = \sum_{n\ge 0}\left\{x_n^3 T_m(x_n) - c_m\frac{\log^{m-2}x_n}{x_n}\right\} + c_m\gamma_{m-2}(a).} \tag{6.19}$$

The displayed series converges absolutely and locally uniformly in $a > 0$ and the masses. Also $\Delta_1(x^3) = \delta^2/8$.

Proof. Choose $K = 2$. Here $h_2(s) = \delta^2 s^2/2$ exactly, so $A_{2,2} = \delta^2$ and $A_{2,j} = 0$ for $j > 2$. There is just the single resonant integer $2k - d = 1$. Equations (6.13) and (6.15) give the result. □

For clarity the first two instances are

$$\Delta_2(x^3) = \sum_{n\ge 0}\left\{x_n^3\log^2\frac{x_n^2+B}{x_n^2+C} - \frac{\delta^2}{x_n}\right\} - \delta^2\psi(a), \tag{6.20}$$

$$\Delta_3(x^3) = \sum_{n\ge 0}\left\{-3x_n^3\log((x_n^2+B)(x_n^2+C))\log^2\frac{x_n^2+B}{x_n^2+C} + 12\delta^2\frac{\log x_n}{x_n}\right\} - 12\delta^2\gamma_1(a). \tag{6.21}$$

In (6.20) the summand is $O(x_n^{-3})$; in (6.21) it is $O(x_n^{-3}\log x_n)$. The all-order formula isolates one prescribed generalized Stieltjes constant, instead of introducing an undefined finite part of a logarithmic series.

There is already an additional local correction at the next odd weight. For $W = x^5$ and $r = 2$,

$$\begin{aligned}\Delta_2(x^5) = \sum_{n\ge 0}&\left\{x_n^5\log^2\frac{x_n^2+B}{x_n^2+C} - \delta^2 x_n + \frac{\delta^2(B+C)}{x_n}\right\}\\ &+ \delta^2\zeta(-1,a) - \delta^2(B+C)\gamma_0(a) - \frac{\delta^2(B+C)}{4}.\end{aligned} \tag{6.22}$$

Indeed $A_{3,2} = -\delta^2(B+C)$ and $A_{3,3} = -3\delta^2(B+C)$. The final term in (6.22) comes from the latter coefficient multiplied by the Hurwitz pole. Simply assigning finite parts to the squared logarithm would miss it.

6.3 The source's conjugate-linear factorization problem

The second question in [5] specifically factors a single quadratic into its two conjugate linear factors. This requires a different spectral scale from a product of two quadratics. Here is the explicit specialization, including that scale. For $a, b > 0$ define

$$F_{W,\pm}(s) = \sum_{n\geq 0} W(x_n)(x_n \pm ib)^{-s}, \qquad Z_W(s) = \sum_{n\geq 0} W(x_n)(x_n^2+b^2)^{-s},$$

using principal logarithms. Since $x_n > 0$, $\log(x_n+ib) + \log(x_n-ib) = \log(x_n^2+b^2)$ without a branch term. Put

$$\widehat{\mathfrak{D}}_W(s) = \tfrac{1}{2}\{F_{W,+}(2s) + F_{W,-}(2s)\} - Z_W(s), \qquad \widehat{\Delta}_m = \widehat{\mathfrak{D}}_W^{(m)}(0).$$

The relevant coefficient polynomials are

$$\widehat{h}(s,t) = \tfrac{1}{2}\{(1+ibt)^{-2s} + (1-ibt)^{-2s}\} - (1+b^2t^2)^{-s} = \sum_{k\geq 1} \eta_k(s) t^{2k}, \tag{6.23}$$

$$\eta_k(s) = (-1)^k b^{2k}\left\{\frac{(2s)_{2k}}{(2k)!} - \frac{(s)_k}{k!}\right\}, \qquad \widehat{A}_{k,j} = \eta_k^{(j)}(0). \tag{6.24}$$

Every η_k is divisible by s^2, although its degree can now be $2k$. In particular $\eta_1(s) = -2b^2s^2$.

Corollary 6.4 (Every conjugate-linear spectral defect)**.** *Choose K as in (6.7). The complete formula (6.13) holds for $\widehat{\Delta}_m$ with these replacements: the finite coefficient sums begin at $k=1$; $A_{k,j}$ is replaced by $\widehat{A}_{k,j}$; every spectral scale $2r$ is replaced by 2; and the raw derivative becomes*

$$\widehat{T}_m(x) = (-1)^m\left\{2^{m-1}(\ell_+(x)^m + \ell_-(x)^m) - (\ell_+(x)+\ell_-(x))^m\right\}, \qquad \ell_\pm(x) = \log(x \pm ib). \tag{6.25}$$

In particular the resonant correction is exactly

$$\widehat{\mathcal{C}}_{m,k}(1;a) = \frac{\widehat{A}_{k,m+1}}{2(m+1)} + \sum_{j=2}^{m}\binom{m}{j}\widehat{A}_{k,j}(-2)^{m-j}\gamma_{m-j}(a). \tag{6.26}$$

All remainder sums are ordinary and absolutely convergent.

Proof. The exact finite-subtraction identity is

$$\widehat{\mathfrak{D}}_W(s) = \sum_{n\geq 0} W(x_n)x_n^{-2s}\left\{\widehat{h}(s,x_n^{-1}) - \sum_{k=1}^{K}\eta_k(s)x_n^{-2k}\right\} + \sum_{d=0}^{D} w_d \sum_{k=1}^{K}\eta_k(s)\zeta(2s+2k-d,a).$$

Its tail is normally convergent for $\Re s > (D-2K-1)/2$. The finitely many terms with $x_n \leq b$ are kept exactly, so the binomial expansion is used only within its tail disk. The principal logarithms are analytic because the linear factors stay in the right half-plane. Taylor differentiation of this identity is the proof of Theorem 6.1 with scale 2 and coefficient (6.24); its pole is $1/(2s)$. This proves the stated correction and convergence claims. □

For the first odd weight $W = x$, put $\widehat{c}_m = -4b^2\binom{m}{2}(-2)^{m-2}$ for $m \geq 2$. Then the exact all-order identity simplifies to

$$\widehat{\Delta}_m(x) = \sum_{n\geq 0}\left\{x_n\widehat{T}_m(x_n) - \widehat{c}_m\frac{\log^{m-2}x_n}{x_n}\right\} + \widehat{c}_m\gamma_{m-2}(a), \qquad \widehat{\Delta}_1(x) = -b^2. \tag{6.27}$$

At order two, $\widehat{T}_2(x) = -4\arctan^2(b/x)$, so

$$\boxed{\widehat{\Delta}_2(x) = 4\sum_{n\geq 0}\left\{\frac{b^2}{x_n} - x_n\arctan^2\frac{b}{x_n}\right\} + 4b^2\psi(a).} \tag{6.28}$$

Evenness of W removes the first-derivative local anomaly but does not remove the higher nonlocal part. Already at $W = 1$,

$$\widehat{\Delta}_2(1) = -4\sum_{n\geq 0}\arctan^2\frac{b}{n+a}.$$

The individual linear terms themselves have finite Hurwitz evaluations:

$$F^{(m)}_{W,\pm}(0) = \sum_{d=0}^{D} w_d\sum_{j=0}^{d}\binom{d}{j}(\mp ib)^{d-j}\zeta^{(m)}(-j, a \pm ib). \tag{6.29}$$

Thus the corollary evaluates every $Z^{(m)}_W(0)$ using a finite complex-Hurwitz part and the explicitly convergent remainder. For example,

$$Z''_1(0) = 4\Re\zeta''(0, a+ib) + 4\sum_{n\geq 0}\arctan^2\frac{b}{n+a}. \tag{6.30}$$

Equations (6.26) and (6.25) answer the stated higher-jet factorization problem without assuming that its first-derivative parity criterion persists.

6.4 A digamma integral and an anchored dilogarithm primitive

The cubic-weight second derivative has a compact ordinary integral. For $B > C \geq 0$,

$$\boxed{\Delta_2(x^3) = 2\int_C^B t\,\Re\psi(a+i\sqrt{t})\log\frac{B-t}{t-C}\,dt.} \tag{6.31}$$

The endpoint logarithms are integrable. To prove it, first use

$$\log^2\frac{x^2+B}{x^2+C} = \int_C^B\int_C^B \frac{ds\,dt}{(x^2+s)(x^2+t)}.$$

The normally convergent digamma difference is

$$\sum_{n\geq 0}\left\{\frac{x_n}{x_n^2+t} - \frac{1}{x_n}\right\} = \psi(a) - \Re\psi(a+i\sqrt{t}).$$

Partial fractions first give the subtracted double sum $\psi(a) + (f(s) - f(t))/(t-s)$, where $f(t) = t\Re\psi(a+i\sqrt{t})$. The term $\delta^2\psi(a)$ from integrating its constant part is cancelled by the explicit $-\delta^2\psi(a)$ in (6.20). The remaining kernel is $(f(s) - f(t))/(t-s)$. The diagonal is removable. A symmetric exclusion of $s = t$, followed by elementary integration of $1/(t-s)$, gives $2\int_C^B f(t)\log((B-t)/(t-C))dt$. This proves (6.31) without separating divergent series.

For an explicit primitive in the shift, set $\delta = B - C > 0$, $y = (x^2 + B)/(x^2 + C) > 1$, and define

$$J_2(y) = -\frac{y \log^2 y}{y-1} - 2\,\mathrm{Li}_2(1-y), \tag{6.32}$$

$$J_3(y) = -\frac{\log^2 y}{2(y-1)^2} + \frac{1}{2}\log^2 y + \mathrm{Li}_2(1-y) - \frac{\log y}{y-1} + \log(y-1) - \log y, \tag{6.33}$$

$$\Phi_{B,C}(x) = -\frac{\delta^2}{2} J_3(y) + \frac{C\delta}{2} J_2(y) - \delta^2 \log x + \frac{\delta^2}{2}\log\delta - \frac{3}{4}\delta^2. \tag{6.34}$$

Here $1 - y < 0$, so the dilogarithm is real and unambiguous.

Theorem 6.5 (Convergent dilogarithm–Gamma primitive)**.** *For $a, a_0 > 0$ and $B > C \geq 0$,*

$$\boxed{\begin{aligned}\int_{a_0}^{a} \Delta_2(x^3; u, B, C)\, du = \sum_{n\geq 0} \{\Phi_{B,C}(n+a) - \Phi_{B,C}(n+a_0)\} \\ - \delta^2 \log \frac{\Gamma(a)}{\Gamma(a_0)}.\end{aligned}} \tag{6.35}$$

Even the two sums of Φ separately converge absolutely. Every normalization constant has been fixed in (6.34).

Proof. Direct differentiation using $\mathrm{Li}_2'(z) = -\log(1-z)/z$ gives $J_2'(y) = \log^2 y/(y-1)^2$ and $J_3'(y) = \log^2 y/(y-1)^3$. The substitution $v = x^2 = \delta/(y-1) - C$ consequently proves

$$\Phi_{B,C}'(x) = x^3 \log^2 \frac{x^2 + B}{x^2 + C} - \frac{\delta^2}{x}.$$

As $y = 1 + \varepsilon \downarrow 1$, $J_2(y) \to 0$ and $J_3(y) = \log\varepsilon - 3/2 + O(\varepsilon)$. These expansions show that the last two constants in (6.34) make $\Phi_{B,C}(x) \to 0$ as $x \to \infty$. Since its derivative is $O(x^{-3})$, one has $\Phi_{B,C}(x) = O(x^{-2})$. Its series and its differentiated series are therefore locally normally convergent for $a > 0$. Sum the derivative and use (6.20) together with $(\log\Gamma)' = \psi$. Evaluation at a_0 fixes the anchor and proves the displayed identity. □

The higher Stieltjes terms in (6.19) also have the classical anchored primitives

$$\int_{a_0}^{a} \gamma_n(u)\, du = \frac{(-1)^{n+1}}{n+1}\{\zeta^{(n+1)}(0, a) - \zeta^{(n+1)}(0, a_0)\}. \tag{6.36}$$

This follows by expanding $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ at $s = 0$; it is not a new identity claimed by this article. Equation (6.35) provides the additional explicit dilogarithmic primitive for the convergent nonlinear part at the second derivative.

7 Source audit and proposed corrections

The inspected sources support the preceding developments, but the research-status map needs an additional cross-reference. The correction below concerns attribution and an already answered construction problem. It does not invalidate the same-direction spectral representations. No counterexample to the inspected source reconstruction or quadratic factorization theorems was found.

7.1 The connected double family is an established interpolant

For $\Re q > 0$ write

$$Z_q(s_1,\dots,s_d) = \sum_{n_1>\cdots>n_d\ge 0} \prod_{j=1}^{d} (n_j+q)^{-s_j}, \qquad Z_q(\varnothing) = 1,$$

initially in an absolute-convergence domain. The same-direction report defines

$$\mathcal{C}_{A,B}(q) = \zeta(A,B) - Z_q(A,B) + \zeta(A,q)\{\zeta(B,q) - \zeta(B)\}. \tag{7.1}$$

The earlier independent-orders report defines the relative interpolant by ordered convolution [7]:

$$Z_b(w) = \sum_{w=uv} Z_a(u)\mathcal{H}_{a,b}(v), \qquad \mathcal{H}_{a,b}(\varnothing) = 1. \tag{7.2}$$

The sum is over prefix–suffix decompositions of the word. At depth two,

$$\mathcal{H}_{q,1}(A,B) = \zeta(A,B) - Z_q(A,B) - \zeta(A,q)\{\zeta(B) - \zeta(B,q)\}.$$

Thus there is an exact identity of the full functions,

$$\boxed{\mathcal{C}_{A,B}(q) = \mathcal{H}_{q,1}(A,B) = \Psi(A,B;q-1).} \tag{7.3}$$

The final member is the Ihara–Nakamura–Yamamoto interpolant [10, Proposition 2.1 and Corollary 4.2]. Their theorem already gives entire dependence on all independent indices at arbitrary depth. The relative report explicitly cites that result and develops its two-endpoint form. This identification is algebraic on an open convergence domain, so it does not rely on uniqueness from values at integer q.

For clarity, the established general finite expression is

$$\mathcal{H}_{q,1}(s_1,\dots,s_d) = \sum_{j=0}^{d} (-1)^j Z_q^{\star}(s_j,\dots,s_1)\,\zeta(s_{j+1},\dots,s_d). \tag{7.4}$$

The empty prefix and suffix have value one, and $Z_q^{\star}$ uses weak inequalities. This is the convolution inverse of the strict ordered series. It preserves the published entire normalization, rather than choosing separate finite parts for its meromorphic constituents.

The corresponding divided-difference kernel is also already available [10, 8]. If $X_0 = 0$, $X_j = v_1 + \cdots + v_j$, then an equivalent convenient form is

$$K_q^{(d)}(v_1,\dots,v_d) = (-1)^d [e^{X_0}, e^{X_1}, \dots, e^{X_d}](y \mapsto y^{d-q}). \tag{7.5}$$

Here the brackets are the divided difference in y, extended confluently when nodes coincide. Its normalized Mellin transform is $\mathcal{H}_{q,1}$, initially when every $\Re s_j > 0$. At $d = 2$ it is precisely the kernel of the same-direction report. To compare with the earlier formula using nodes e^{-X_j}, invert each node in its finite barycentric sum: the product of all nodes cancels the exponential prefactor, leaving the sign $(-1)^d$ and the power $d - q$ in (7.5).

The shift law and the depth-two stuffle law in this normalization are

$$\mathcal{H}_{q+1,1}(s_1,\dots,s_d) = \mathcal{H}_{q,1}(s_1,\dots,s_d) + q^{-s_1}\mathcal{H}_{q,1}(s_2,\dots,s_d), \tag{7.6}$$

$$h_q(s)h_q(t) = \mathcal{H}_{q,1}(s,t) + \mathcal{H}_{q,1}(t,s) + h_q(s+t), \quad h_q(s) = \zeta(s) - \zeta(s,q). \tag{7.7}$$

They are inherited identities for entire functions. Sign conventions can be checked immediately: the same-direction report uses the depth-one difference $\zeta(s,q) - \zeta(s) = -h_q(s)$, which accounts for the apparent change of sign in its displayed laws.

Proposed source edit. The attribution paragraph of [6] should identify its entire connected double family as the depth-two specialization of [10, 7]. Its second further-research question should no longer ask for existence, entireness, shift/stuffle laws, or the basic higher divided-difference kernel. Those constructions are already available. A useful replacement asks for explicit minimal coordinate reductions on specified mixed nonpositive-index loci, or for comparisons between genuinely different normalizations. The Barnes and beta–Newton same-direction representations remain separate results and are not affected by this correction.

7.2 Updates justified by the new proofs

The order-eight statement in [2] correctly retained the congruence qualification $b \not\equiv 1 \pmod{16}$. It was not a false theorem. After the present proof and certificate are accepted, the collective statement may be strengthened to every $b \geq 5$ and its Q1 closed. The original congruence proof remains useful as the finite range reduction. Its previously rejected simple valuation extrapolation should remain rejected; the new proof does not restore that formula.

The same report's Q5 is now answered for the fourth power and its regular even moments. The statement should refer to the six displayed formal coordinates and the all-index Bernoulli transform. It should not assert independence of the evaluated constants or promote fifth and higher powers to the same status. The starting value $V_0 = Q_7$ must retain Hoffman's attribution.

For [5], Q7 is answered by the pairwise logarithmic correction and the confluent resolvent formulas, while Q2 is answered by Corollary 6.4. The latter uses the source's linear factors and spectral scale two. A product of two quadratic factors has scale four; confusing them changes every higher correction. The explicit primitive in Theorem 6.5 is progress on the mixed primitive direction, not a classification of all shift–mass integrals.

7.3 Corrections that must accompany higher-order reuse

Three points are essential when extending the existing formulas. First, the identity $\mathcal{H}_a(-m) = 0$ does not imply $\mathcal{H}'_a(-m) = 0$. At bilateral third order the omitted term would be exactly $-12(B - C)^2 \log(2 \sin \pi a)$ for $P(v) = v^2$. Second, multiplication by a Hurwitz pole uses one more coefficient of the vanishing factor than multiplication by its regular part. The term $A_{k,m+1}/(2r(m + 1))$ in (6.15) is therefore required. Its first additional second-order example is $-\delta^2(B + C)/4$ in (6.22). Third, an antiderivative of a summand is not automatically summable: the constant in $\Phi_{B,C}$ is fixed so that it decays to zero. Without it, the separate series of primitives would diverge.

These are requirements of the new extensions, not errors falsely attributed to source theorems that only treated first derivatives. The article and integration notes retain this distinction.

7.4 Unchanged arithmetic conjectures

The canonical Gaussian S_6 identity and the revised S_8 candidate remain conjectural in this continuation [1]. The rejected older S_8 vector remains different from the revised candidate. No failed relation search, small residual, or finite formal rank calculation is converted here into an arithmetic proof or a nonreduction assertion. Likewise, no new claim is made about the transcendence or numerical independence of the six quartic coordinates. These boundaries should remain visible when the results are integrated into the collective text.

8 Further research questions with exact targets

The following problems are organized around the structures proved above. Each asks for an identity, a finite reduction, or a proof of a specified conjecture. A successful result should identify the precise function class and normalization before a computation is interpreted.

R1. Complete every fixed-order reconstruction family. Theorem 3.1 supplies a finite decision procedure once a is fixed. For $a = 5$, the exact threshold $B_5 = 1920$ reduces the smaller-order-ten problem to $6 \le b < 1920$; the adjacent case $b = 6$ and the gap-four case $b = 7$ are already proved. Complete the remaining cases using exact arithmetic, then apply the same procedure at $a = 6$, corresponding to smaller order twelve. A structural factorization or a congruence may eliminate most of the finite range before direct determinants are formed. The target is a complete all-b theorem for each family, with every exceptional index accounted for, rather than a larger table of experiments.

R2. Remove the fixed-order qualification. The leading positive determinant gives eventual nonvanishing but does not control the lower Bernoulli degrees uniformly at small separation. Find a determinant identity, an arithmetic invariant, or a recurrence which prevents cancellation for all $a < b$. The existing adjacent and gap-four cases, together with the complete small-order families, provide exact consistency constraints. A proof uniform in a would settle the remaining unrestricted optimal-prefix conjecture.

R3. Give closed coefficient rows for higher-order reconstruction. The ordinary identities (2.21) use a completely specified rational inverse matrix. Seek a product formula or a fixed-size kernel for its rows, comparable to the known smaller-order inverse construction. It should specify every polynomial coefficient directly and retain all subtraction constants. A formal reference to an unevaluated inverse or to unknown minors is a weaker endpoint than such a closed kernel.

R4. Find the fifth- and higher-power tail cancellation. For each integer $d \ge 5$, continue $\sum_{n\ge0} \psi'(n+1)^d x_n^{-s}$ on its regular parity lattice. The quartic boundary identity (4.28) suggests combining the parity components of successive powers before applying finite summation. Determine which lower tail powers survive, identify a fixed coordinate space for each d, and construct its anchored generator. The dimension and the required weights should be proved, not inferred from the first few moments.

R5. Treat the resonant parity as a separate spectral family. For the quartic family, the points $s = -3 - 2k$ are actual poles with explicit residues h_k. Determine their Laurent constants and higher spectral coefficients by a normalization compatible with (4.4). These coefficients will involve logarithmic moments, so the regular even-moment theorem cannot simply be differentiated at its discrete values. An anchored analytic generator for the Laurent family would be a natural next object.

R6. Determine the functional depth of the quartic generator. The present construction gives a finite depth-at-most-three expression. Specify allowable transformations of arguments and products, then ask whether the particular depth-three combination reduces further. A symbolic relation must be checked by a functional derivative identity and an anchored endpoint. Formal symbol or shuffle rank should not be interpreted as arithmetic irreducibility of a value.

R7. Extend exact reconstruction to mixed and shifted tails. Replace the identical fourth power by $\prod_j \zeta(s_j, n + a_j)$, beginning with positive integer orders and a common rational shift. Determine which parity cancellations survive and the finite replacement for $\zeta(-2m, 1/2) = 0$. Character twists should be handled only after the shift-dependent counterterms are explicit. The goal is ordinary convergent identities with a specified finite arithmetic coordinate set.

R8. Reduce the spectral logarithmic quadrature on special loci. For rational a and selected algebraic masses, seek finite reductions of (5.15) and (6.31) into polylogarithm order derivatives, Gamma derivatives, or a precisely defined multiple-Gamma family. The

ordinary integral is already an exact answer at arbitrary parameters. A finite special-function reduction is an additional theorem and should not be assumed from the first-derivative factorization.

R9. Classify noninteger-weight resonance crossings. Replace a polynomial monomial by x^ρ in the one-sided master. As ρ crosses an odd integer, an argument of the Hurwitz function passes through one. Construct a joint Laurent formula in s and ρ which specializes to (6.15) and distinguishes different orders of taking limits. The local coefficient at each crossing must match the polynomial theorem exactly.

R10. Build primitive formulas at every spectral order. The primitive (6.35) treats the nonlinear second jet with a completely fixed dilogarithmic anchor. Higher logarithmic powers lead naturally to multiple polylogarithms after the same rational substitution. Develop a finite integration rule for each m and combine it with the classical primitives of $\gamma_{m-2}(a)$. Also determine when differentiation in a mass commutes with an anchored shift primitive without an additional endpoint term.

R11. Project the Stieltjes identities to arithmetic characters. At $a = p/q$, apply finite character sums to the all-order correction formulas. Give an exact dictionary between the resulting combinations of $\gamma_j(p/q)$, Hurwitz derivatives, and Dirichlet-L derivatives. The reflected even and odd channels should be retained separately. The first useful deliverable would cover small denominators with the full logarithmic normalization in q.

R12. Return to the Gaussian period identities with new functional input. The unresolved S_6 and revised S_8 identities require a relation that survives the known restrictions of the explored Cayley relation systems. The higher-depth polylogarithms appearing in the quartic generator offer additional functional expressions, but no connection to the target period is established here. A useful next result would give an explicit map from one such functional identity to the exact canonical residual, including all lower-weight and branch terms.

R13. Formalize the finite and analytic interfaces. The Newton triangularization, Bernoulli modular recurrence, finite summation identities, and resonant coefficient extraction are suitable finite proof-assistant targets. The separate analytic obligations are normal convergence of the subtracted families and their agreement with the original series on a common half-plane. A formalization should state those interfaces explicitly. Validated interval evaluation of the remaining quadratures is a different project from formal proof and can be added independently.

A Reproduction, certificates, and evidence boundaries

The package contains the modular article source, a standalone combined TEX source, the compiled PDF, exact coefficient data, the inherited order-eight certificate with provenance, and independent Python verification programs. All mathematical verification programs operate without a network connection. They require Python 3, SymPy, and mpmath; the article builds with a standard modern LATEX installation.

A.1 What the finite certificates establish

The order-eight verifier reconstructs the source four-dimensional matrix from finite differences, clears denominators, and checks its determinant by the defining 24 permutation products. It replays 3,852 exact polynomial coefficient comparisons in the inherited certificate. The new leading Hankel factor is checked both by coefficient extraction and by its triangular structural

factor. The constants K, V, and C, the exact factorial comparison at $b = 512$, and all 31 remaining indices are evaluated exactly. The finite indices are checked by both rational Bernoulli numbers and a separate recurrence modulo 2^{32}.

The all-b conclusion does not follow from a finite sampling rule. It follows from the analytic domination proof, the inherited congruence reduction, and the exhaustive finite residual set. The general fixed-order theorem uses its constructive factorial comparison to prove termination; the example thresholds do not stand in for that termination argument.

Ten additional exact normalization comparisons use $a = 1, \dots, 5$ and $b = 2a, 2a + 1$. They independently compare the original matrix, its shifted Hankel block, the finite Padé sums, and the retained weighted-Bernoulli matrix. The separate threshold script computes every coefficient norm used in the table in Section 3 by exact arithmetic.

The artifact `order8_identity_b17_k0.json` specifies all integer coefficients of the ordinary convergent extraction identity for $(p, q) = (8, 34)$. Its generator verifies the coordinate cancellation against the original matrix, so the example does not depend on transcribing a long printed coefficient list.

The quartic verifier reconstructs the finite-cutoff coordinates through Faulhaber's formula, independently of the closed D_j expression. A separate exact shuffle, stuffle, and duality calculation reduces the multiple zeta values through weight seven. The replay checks 25 quartic coordinates, 20 centered moments, 24 boundary identities, 24 generator coefficients, the six-coordinate determinant, and the two coefficient identities behind the rational extraction.

The bilateral spectral verifier compares rational Pochhammer coefficients with coefficients constructed directly from powers of elementary logarithmic series. It checks 798 exact rational identities, including several numbers of masses. The one-sided verifier adds 142 exact symbolic identities, including the general Hurwitz Laurent product, the additional local pole term, both dilogarithm derivatives, the primitive normalization, and the conjugate-linear coefficients.

The columns of the first six quartic rows correspond, in order, to Q_7, Q_5, $\zeta(2)$, $\zeta(3)$, $\zeta(4)$, and 1:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{12} & \frac{1}{3} & 0 & 6 & -\frac{17}{6} & 0 \\ \frac{7}{240} & -\frac{1}{6} & -\frac{1}{10} & -\frac{3}{2} & \frac{17}{30} & -\frac{1}{10} \\ -\frac{31}{1344} & \frac{7}{48} & \frac{1}{168} & \frac{13}{20} & \frac{17}{672} & \frac{11}{280} \\ \frac{127}{3840} & -\frac{31}{144} & \frac{299}{2160} & -\frac{2551}{5040} & -\frac{289}{540} & -\frac{17}{840} \\ -\frac{2555}{33792} & \frac{127}{256} & -\frac{3667}{6336} & \frac{45929}{67200} & \frac{497267}{253440} & \frac{1187}{277200} \end{pmatrix}$$

Their exact determinant is $7361/49896000$.

A.2 Independent numerical diagnostics

The numerical work tests different representations, with every working precision and cutoff stored in its JSON record. The quartic checks use 120 decimal working digits and two finite-head/centered-Hurwitz-tail choices, $(N, J) = (120, 36)$ and $(240, 54)$. For $0 \le r \le 8$, the largest residual from the second choice is below 3.6×10^{-90}.

The bilateral calculations use 75 decimal working digits. The spectral continuations and logarithmic tails are constructed separately; the elementary resolvent quadrature and confluent expression provide additional comparisons. In particular, at $a = 1/2$ and masses $4, 1$,

$$S_1 = 4.25062684984826912051041637407031330017\ldots,$$

whereas at $P(v) = v^2$,

$$S_{v^2} = -66.37570183951562430903381244813937818\ldots.$$

These values illustrate that the subtraction and the polynomial weight are part of the identity; the sign of the unweighted logarithmic square is not a sign rule for every subtracted weighted sum.

The one-sided diagnostics use 85 decimal working digits and 22 strict comparisons, with tolerance 10^{-60}. Individual spectral families use 11 head terms and 72 tail coefficients; the elementary logarithmic sums use 12 head terms and 64 coefficients. The largest recorded strict difference is below 2×10^{-71}. The choices include $(a, B, C) = (1/3, 9, 0)$, for which $B/a^2 = 81$; thus the implementation does not depend on an invalid global small-mass expansion. Both orientations of the anchored primitive are compared with a separate Gamma quadrature. Two further large-x checks test its asymptotic normalization.

None of these decimal diagnostics is an interval certificate or a proof of an identity. They test the independently derived formulas and their implementation. The proofs of meromorphic continuation, normal convergence, and all-index cancellation are in the article. No proof-assistant formalization is supplied.

A.3 Entry points and integration

Run `pythonverification/run_all.py` to execute the default mathematical suites in isolated subprocesses. The exact order-eight source replay is included in that default. The scripts also run individually in their respective directories. The recorded receipts identify their actual runtime versions and completion status.

Build the article with `latexmk-pdf-interaction=nonstopmode-halt-on-errorarticle.tex`. Alternatively compile `article_singlefile.tex` twice with pdfLaTeX. The combined source embeds every section and bibliography and has no external mathematical-input dependency.

The integration notes specify the source claims to update, the theorem boundaries to retain, and suggested manuscript destinations. Historical incoming reports need not be overwritten: the new proof can be added as a continuation with explicit cross-references. The machine-readable `CLAIM_STATUS.json` separates new proved claims, inherited inputs, editorial corrections, and unresolved questions. The final `SHA256SUMS` covers all distributed source, code, data, and PDF files other than that manifest itself. Rebuilding may alter PDF metadata; a new build should receive its own rendered review.

References

[1] V. Reshetnikov and ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, collective manuscript, four volumes and associated research reports. Inspected source revision `73f18ef013d5b74eafc883efb2c15210642df12a`, 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/73f18ef013d5b74eafc883efb2c15210642df12a/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt research collection, *Exact Reconstruction and Higher Resonances in Polygamma–Zeta Calculus*, incoming research report, 11 October 2026. In particular Sections 2, 5, 6, and 9. Archive `ProveIt_Polygamma_Reconstruction_and_Resonance_2026-10-11.zip` in `docs/incoming`, at `73f18ef013d5b74eafc883efb2c15210642df12a`. https://github.com/VladimirReshetnikov/ProveIt/tree/73f18ef013d5b74eafc883efb2c15210642df12a/docs/incoming.

[3] ProveIt research collection, Bernoulli–Tornheim–digamma continuation, *The general Hankel reduction*, source section `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/sections/03b_general_hankel.tex`, at `73f18ef013d5b74eafc883efb2c15210642df12a`. The general small determinant and Padé row reduction are reproduced in Section 3.

[4] ProveIt Contributors, *Centered trigamma moments*, manuscript source `Analysis/Polylogarithms/docs/manuscript/chapters/07-centered-trigamma-moments.tex`, at `73f18ef013d5b74eafc883efb2c15210642df12a`. The all-even square moment formula and its $\mathrm{Li}_2/\mathrm{Li}_3/\mathrm{Li}_4$ generator.

[5] ProveIt research collection, *Quadratic-Spectrum Stieltjes Identities*, incoming research report, 10 October 2026. In particular Sections 5–7 and further-research questions 2 and 7. Archive `ProveIt_Quadratic_Spectrum_Stieltjes_2026-10-10.zip` in `docs/incoming`, at `73f18ef013d5b74eafc883efb2c15210642df12a`.

[6] ProveIt research collection, *Same-Direction Polylogarithm Products*, incoming research report, 10 October 2026. Connected coordinates, same-direction representations, attribution paragraph, and further-research question 2. Archive `ProveIt_Same_Direction_Complex_Orders_2026-10-10.zip` in `docs/incoming`, at `73f18ef013d5b74eafc883efb2c15210642df12a`.

[7] ProveIt research collection, independent-orders continuation, *An entire normalization for all independent harmonic exponents*, source section `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/independent-orders/sections/02_hurwitz.tex`, at `73f18ef013d5b74eafc883efb2c15210642df12a`. The relative interpolant, its exact identification with the published interpolant, and the endpoint and stuffle laws.

[8] ProveIt research collection, harmonic-gaps and mixed-moments continuation, *A jointly entire kernel for all zero-block decorations*, source section `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/harmonic-gaps-mixed-moments/sections/02_joint_generator.tex`, at `73f18ef013d5b74eafc883efb2c15210642df12a`. In particular the established divided-difference kernel and its normalized Mellin transform.

[9] M. E. Hoffman, *Sums of products of Riemann zeta tails*, Mediterranean Journal of Mathematics **13** (2016), 2771–2781. https://doi.org/10.1007/s00009-015-0653-9. Preprint https://arxiv.org/abs/1602.03399. The fourth-power evaluation is printed immediately after Corollary 4.4.

[10] K. Ihara, Y. Nakamura, and S. Yamamoto, *Interpolant of truncated multiple zeta functions*, The Ramanujan Journal **67** (2025), article 1. Published 4 March 2025. https://doi.org/10.1007/s11139-025-01045-2.

[11] E. Elizalde, L. Vanzo, and S. Zerbini, *Zeta-function regularization, the multiplicative anomaly and the Wodzicki residue*, preprint (1997). https://arxiv.org/abs/hep-th/9701060.

[12] E. Gürel, *Resolution of multiplicative anomaly of zeta regularization for polynomials*, arXiv:2504.14563, version 2, 3 September 2025. https://arxiv.org/abs/2504.14563.

[13] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*. In particular Sections 5.5 and 5.15 (Gamma and polygamma), 24.8 (Bernoulli Fourier expansions), and 25.2, 25.11, 25.12 (zeta, Hurwitz zeta, and polylogarithms). Accessed 11 October 2026. https://dlmf.nist.gov/.